MILITARY SPECIFICATION

# MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE

# MONOLITHIC 16-BIT MICROPROCESSOR (FIXED INSTRUCTION)

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic NMOS, silicon gate microprocessor. Three product assurance classes are provided.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

| Device type | Circuit |
|---|---|
| 01 ($T_C$ = 125°C max) | 16-bit, fixed instruction microprocessor |

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

| Letter | Case outline, (see MIL-M-38510, appendix C) |
|---|---|
| Q | D-5 (40-lead, 9/16" x 2-1/16" dual-in-line pack) |

1.3 Absolute maximum ratings.

| | |
|---|---|
| $V_{CC}$ supply voltage range ($V_{CC}$ - GND) | -1.0 to +7.0 V |
| Input voltage range | |
| Storage temperature range | -65° to +150°C |
| Power dissipation for package | 2.5 W |
| Power dissipation for device type 01: | |
| $T_C$ = -55°C | 2.3 W |
| $T_C$ = 125°C | 1.6 W |
| Lead temperature (soldering - 5 seconds) | 270°C |
| Maximum junction temperature at $T_C$ = 125°C | $T_j$ = 150°C |
| Thermal resistance, junction to case | 15°C/W |

1.4 Recommended operating conditions.

| | |
|---|---|
| Supply voltage $V_{CC}$ | +4.5 to +5.5 V |
| High-level input voltage | |
| Logic inputs, ($V_{IH1}$) | 2.2 V to ($V_{CC}$ +0.5 V) |
| Clock input, ($V_{IH2}$) | 3.9 V to ($V_{CC}$ +1.0 V) |
| Low-level input voltage: | |
| Logic inputs, ($V_{IL1}$) | -0.5 V to 0.6 V |
| Clock input, ($V_{IL2}$) | -0.5 V to 0.6 V |
| Frequency of operation | 2 MHz to 5 MHz |
| Case operating temperature range | -55°C to +125°C |
| Clock rise time (from 1.0 V to 3.5 V) | 10 ns |
| Clock fall time (from 3.5 V to 1.0 V) | 10 ns |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification For

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.5.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.6).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended case operating temperature range unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in table II for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.6 Microprocessor instructions. Microprocessor instructions shall be as specified in table V.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group covering only NMOS microprocessors.



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TABLE I. Electrical performance characteristics.

| Test | Symbol | Conditions | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|
| $V_{CC}$ = 5.0 V ±10%, $T_C$ = -55°C to 125°C unless otherwise specified | | | | | |
| Input low voltage (except CLK) | $V_{IL1}$ | | -0.5 | +0.6 | V |
| Input high voltage (except CLK) | $V_{IH1}$ | | 2.2 | $V_{CC}$+0.5 | V |
| Clock input low voltage | $V_{IL2}$ | | -0.5 | +0.5 | V |
| Clock input high voltage | $V_{IH2}$ | | 3.9 | $V_{CC}$+1.0 | V |
| Output low voltage | $V_{OL}$ | $I_{OL}$ = 2.0 mA, $V_{CC}$ = 5.5 V Figure 5 LOAD | | 0.45 | V |
| Output high voltage | $V_{OH}$ | $I_{OH}$ = 400 A, $V_{CC}$ = 4.5 V Figure 5 LOAD | 2.4 | | V |
| Power supply current | $I_{CC}$ | $T_C$ = -55°C $V_{CC}$ = 5.5 V | | 340 | mA |
| Input low current (NMI, INTR, CLK, RESET, READY, $\overline{TEST}$, MN/$\overline{MX}$) | $I_{IL1}$ | $V_{IN}$ = 0.0 V, $V_{CC}$ = 5.5 V | -10 | 10 | µA |
| Input high current (NMI, INTR, CLK, RESET, READY, $\overline{TEST}$, MN/$\overline{MX}$) | $I_{IH1}$ | $V_{IN}$ = 5.5 V, $V_{CC}$ = 5.5 V | -10 | 10 | µA |
| Input low current ($AD_{0-15}$) | $I_{IL2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.45 V | -10 | 10 | uA |
| Input high current ($AD_{0-15}$) | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | -10 | 10 | µA |
| High impedance state leakage current (A/S, $\overline{BHE}$/$S_7$, $\overline{RD}$, DEN($\overline{S_0}$), DT/$\overline{R}$($S_1$), M/$\overline{IO}$($\overline{S_2}$), $\overline{WR}$($\overline{LOCK}$)) | $I_{ZL1}$ | $V_{CC}$ = 5.5 V, $V_O$ = 0.45 V | -10 | +10 | µA |
| | $I_{ZH1}$ | $V_{CC}$ = 5.5 V, $V_O$ = 2.4 V | -10 | +10 | µA |
| Capacitance of input buffer (all input except $AD_0$-$AD_{15}$, $\overline{RQ}$/$\overline{GT}$) | $CI_1$ | | | 15 | pF |
| Capacitance of I/O buffer ($AD_0$-$AD_{15}$, $\overline{RQ}$/$\overline{GT}$) | $CI_2$ | | | 15 | pF |



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| $V_{CC}$ = 5.0 V ±10%, $T_C$ = -55°C to 125°C unless otherwise specified | | | | | |
| Capacitance of three state outputs (A/S, $\overline{BHE}/S_7$, $\overline{RD}$, $\overline{DEN}(\overline{S_0})$, $DT/\overline{R}(S_1)$, $M/\overline{IO}(S_2)$, $\overline{WR}(\overline{LOCK})$) | CO | | | 15 | pF |
| CLK cycle period | $t_{CYC}$ | Figure 6 LOAD | 200 | 500 | ns |
| CLK low time | $t_{PWL1}$ | | $2/3(t_{CYC})-15$ | | ns |
| CLK high time | $t_{PWH1}$ | | $1/3(t_{CYC})+2$ | | ns |
| Data set up time to CLK ↓ EDGE | $t_{SZH}$<br>$t_{SZL}$ | | 30 | | ns |
| Data in hold time from CLK ↓ EDGE | $t_{HHZ}$<br>$t_{HLZ}$ | | 10 | | ns |
| Ready setup time to CLK ↓ EDGE | $t_{SLH1}$ | | $2/3(t_{CYC})-15$ | | ns |
| Ready hold time from CLK ↓ EDGE | $t_{HHL1}$<br>$t_{HLH1}$ | | 30 | | ns |
| Request ($\overline{RQ}/GT$) hold time from CLK ↓ EDGE | $t_{HLH2}$ | | 40 | | ns |
| Request ($\overline{RQ}/GT$) setup time to CLK ↓ EDGE | $t_{SHL2}$ | | 30 | | ns |
| Ready active to status passive. Applies to T3 and WAIT states ($S_0$-$S_2$) | $t_{PLH1}$ | | | 110 | ns |
| Status active delay from CLK ↓ EDGE ($S_{0-2}$) | $t_{PHL2}$ | | 10 | 110 | ns |
| Status inactive delay from CLK ↓ EDGE ($S_{0-2}$) | $t_{PLH2}$ | | 10 | 130 | ns |
| Address valid delay from CLK ↓ EDGE ($AD_{0-15}$, A16/$S_3$-A19/$S_6$) | $t_{PHL3}$<br>$t_{PLH3}$ | | 10 | 110 | ns |



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| $V_{CC}$ = 5.0 V ±10%, $T_C$ = -55°C to 125°C unless otherwise specified | | | | | |
| Control valid delay from CLK ↓ EDGE ($\overline{BHE}/S_7$, QS1,0,LOCK) | $t_{PHL4}$<br>$t_{PLH4}$ | Figure 6 load | 10 | 110 | ns |
| Address invalid delay from CLK ↓ EDGE ($AD_{0-15}$,A16/$S_3$-A19/$S_6$) | $t_{PHL5}$<br>$t_{PLH5}$ | | 10 | | ns |
| $\overline{BHE}/S_7$ invalid delay from CLK ↓ EDGE | $t_{PHL6}$<br>$t_{PLH6}$ | | 10 | | ns |
| Address float delay from CLK ↓ EDGE | $t_{PHZ1}$<br>$t_{PLZ1}$ | | $t_{PHL5}$ | 80 | ns |
| Data valid delay from CLK ↓ EDGE ($AD_{0-15}$) | $t_{PHL7}$<br>$t_{PLH7}$ | | 10 | 110 | ns |
| Status valid delay from CLK ↓ EDGE ($\overline{BHE}/S_7$, A16/$S_3$,A19/$S_6$) | $t_{PHL8}$<br>$t_{PLH8}$ | | 10 | 110 | ns |
| Data invalid delay from CLK ↓ EDGE ($AD_{0-15}$) | $t_{PHZ2}$<br>$t_{PLZ2}$ | | 10 | 110 | ns |
| Status invalid delay from CLK ↓ EDGE ($\overline{BHE}/S_7$, A16/$S_3$-A19/$S_6$) | $t_{PHL9}$<br>$t_{PLH9}$ | | 10 | | ns |
| Address float to read active | $t_{PHL22}$ | | 0 | | ns |
| $\overline{RD}$ active delay from CLK ↓ EDGE | $t_{PHL10}$ | | 10 | 165 | ns |
| $\overline{RD}$ inactive delay from CLK ↓ EDGE | $t_{PLH10}$ | | 10 | 150 | ns |
| $\overline{RD}$ inactive to next address active | $t_{PZH}$<br>$t_{PZL}$ | | $t_{CYC}$-45 | | ns |
| Grant ($\overline{RQ}/\overline{GT}$) active delay from CLK ↓ EDGE | $t_{PHL11}$ | | 0 | 85 | ns |
| Grant ($\overline{RQ}/\overline{GT}$) inactive delay from CLK ↓ EDGE | $t_{PLH11}$ | | 0 | 85 | ns |



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| $V_{CC}$ = 5.0 V ±10%, $T_C$ = -55°C to 125°C unless otherwise specified | | | | | |
| $\overline{RD}$ WIDTH | $t_{PWL2}$ | Figure 6 load | $2t_{CYC}-75$ | | ns |
| $\overline{WR}$ width | $t_{PWL3}$ | | $2t_{CYC}-60$ | | ns |
| Hold setup time to CLK ↓ EDGE | $t_{SLH3}$<br>$t_{SHL3}$ | | 35 | | ns |
| INTR, NMI, $\overline{TEST}$ setup time setup req. for asynchronous signal only to CLK ↓ EDGE | $t_{SHL4}$<br>$t_{SLH4}$ | | 30 | | ns |
| ALE width | $t_{PWH4}$ | | $t_{PWL1}-20$ | | ns |
| ALE active delay from CLK ↓ EDGE | $t_{PLH12}$ | | 0 | 80 | ns |
| ALE inactive delay from CLK ↓ EDGE | $t_{PHL12}$ | | | 85 | ns |
| ALE inactive to $AD_0$-$AD_{15}$ inactive (RD cycle) | $t_{PHZ3}$<br>$t_{PLZ3}$ | | $t_{PWH1}-10$ | | ns |
| ALE inactive to $\overline{BHE}/S_7$, $A19/S_6$-$A16/S_3$ inactive | $t_{PHL13}$<br>$t_{PLH13}$ | | $t_{PWH1}-10$ | | ns |
| Address valid to ALE low ($AD_{0-15}$, $A16/S_3$-$A19/S_6$) | $t_{PHL14}$ | | $t_{PWL1}-60$ | | ns |
| $\overline{BHE}/S_7$ valid to ALE low | $t_{PHL15}$ | | $t_{PWL1}-60$ | | ns |
| Data inactive to $\overline{WR}$ inactive ($AD_{0-15}$) | $t_{PHZ4}$<br>$t_{PLZ4}$ | | $t_{PWL1}-30$ | | ns |
| $\overline{WR}$, $\overline{INTA}$, active delay from CLK ↓ EDGE | $t_{PHL16}$ | | 10 | 110 | ns |
| $\overline{DT/R}$, M/$\overline{IO}$ active delay from CLK ↓ EDGE | $t_{PHL17}$<br>$t_{PLH17}$ | | 10 | 110 | ns |
| Control inactive delay 1 from CLK ↓ EDGE ($\overline{DEN}$, $\overline{INTA}$, $\overline{WR}$) | $t_{PLH23}$ | | 10 | 110 | ns |



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TABLE I. Electrical performance characteristics - Continued.

| Test | Symbol | Conditions | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| $V_{CC}$ = 5.0 V ±10%, $T_C$ = -55°C to 125°C unless otherwise specified | | | | | |
| Control inactive delay 2 from CLK ↓ EDGE ($\overline{DEN}$ during write cycle) | $t_{PLH24}$ | Figure 6 load | 10 | 110 | ns |
| HLDA valid delay from CLK ↓ EDGE | $t_{PHL18}$<br>$t_{PLH18}$ | | 10 | 160 | ns |
| ALE inactive to $AD_0$-$AD_{15}$ inactive ($\overline{WR}$ cycle) | $t_{PHL19}$<br>$t_{PLH19}$ | | $t_{PWH1}$-10 | | ns |
| $\overline{DEN}$ active delay from CLK ↓ EDGE | $t_{PHL20}$ | | 10 | 110 | ns |
| Address delay 2/ from CLK ↓ EDGE | $t_{PLH21}$<br>$t_{PHL21}$ | | $t_{PHL5}$ | 80 | ns |
| Ready inactive setup time to CLK (applies to T2 state only) | $t_{SHL1}$ | | -8 | | ns |

1/ All timing equations are in nanoseconds.
2/ This delay is for lines $AD_0$-$AD_{15}$. $A19/S_6$-$A16/S_3$, $S_0$-$S_2$, $\overline{RD}$, $\overline{LOCK}$, and $\overline{BHE}/S_7$ during request/grant and HOLD/HLDA sequences only.



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TABLE II. Electrical test requirements.

| MIL-STD-883 test requirement | Subgroups (see table III unless otherwise specified) | | |
|---|---|---|---|
| | Class S devices | Class B devices | Class C devices |
| Interim electrical parameters (pre burn-in) (method 5004) | 1,7 | 1,7 | None |
| Final electrical test parameters (method 5004) | *1,2,3,7,8,9, 10,11 | *1,2,3,7,8,9, 10,11 | 1,2,3,7,9 |
| Group A test requirements (method 5005)(Note 1) | 1,2,3,**4,7,8, 9,10,11 | 1,2,3,**4,7,8, 9,10,11 | 1,2,3,7,9 |
| Group B zap test | See 4.5.3 | See 4.5.3 | See 4.5.3 |
| Group C end-point electrical parameters (method 5005) | None | 2,3,8 | 2,8 (125°C only) |
| Additional electrical subgroups for group C periodic inspection | None | None | None |
| Group b end-point electrical parameters (method 5005) | 1,2,3,7,8 | 2,3,8 | 2,8(125°C only) |

*PDA applies to subgroups 1 and 7 .
**Subgroup 4 ($C_I$, $C_0$) - Qualification only (see 4.4.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. The burn-in test shall be as described in 3.1.10 of method 5004 of MIL-STD-883 and shall be performed as indicated. (3.1.9, 3.1.11, and 3.1.12 are not required.)

2. The circuit shown on figure 4 shall be used. The test temperature shall be 125°C minimum.

b. Percent defective allowable (PDA) - The PDA is specified as 5 percent for class S devices and 10 percent for class B devices based on failures from group A, subgroups 1 and 7 tests after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7 after burn-in, divided by the total number of devices submitted for burn-in in that lot, shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

c. Constant acceleration shall be in accordance with test condition B of method 2001 of MIL-STD-883.



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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II.

b. Subgroup 4 ($C_O$, $C_I$ measurement) shall be performed only for initial qualification and after process or design changes which may affect device capacitance.

c. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

d. Variables data shall be taken for subgroups 9, 10, and 11 only during initial qualification and after process or design change.

e. Subgroups 7 and 8 functional testing shall include verification of the instruction set, table V.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883 and as follows:

a. An additional subgroup shall be added to the group B inspection requirements for class S, B, and C devices and shall consist of the test, conditions, and limits specified in 4.5.3. The LTPD for this subgroup shall be 15 for all classes.

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II.

b. Operating life test (method 1005 of MIL-STD-883).

1. Use life test condition D using the circuit shown on figure 6.

2. Case temperature shall be 125°C.

c. The constant acceleration shall be condition B of test method 2001, MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883;

a. End-point electrical parameters shall be as specified in table II herein.

b. The constant acceleration shall be condition B of test method 2001, MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows:

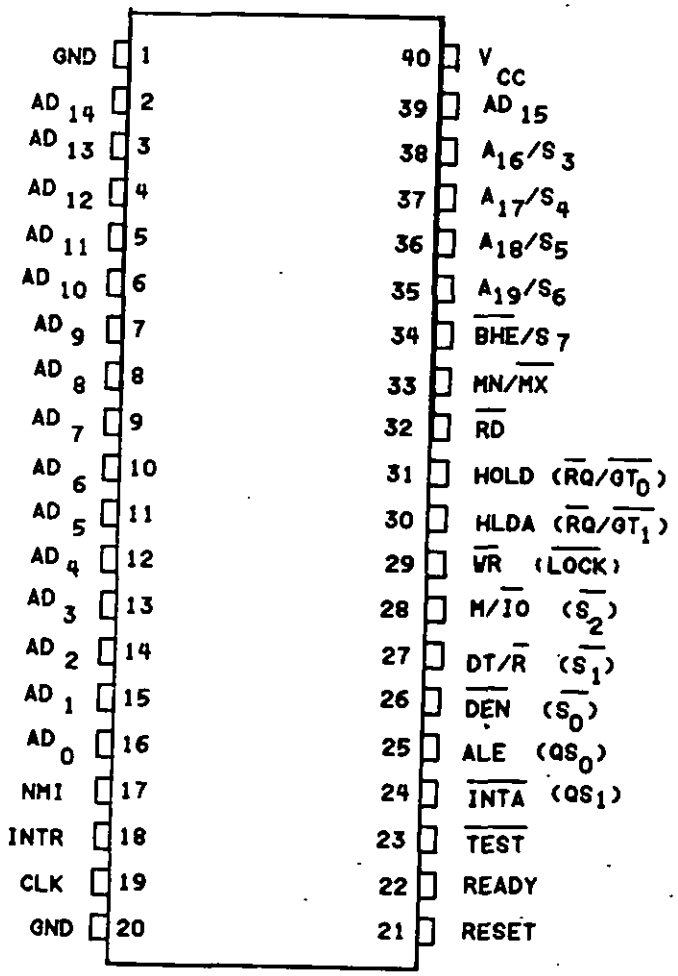
4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life-test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to 35°C prior to removal of the bias.



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Device type 01




Top view

Maximum mode pin functions (e.g., $\overline{LOCK}$) are shown in parentheses.

FIGURE 1. Terminal connections.



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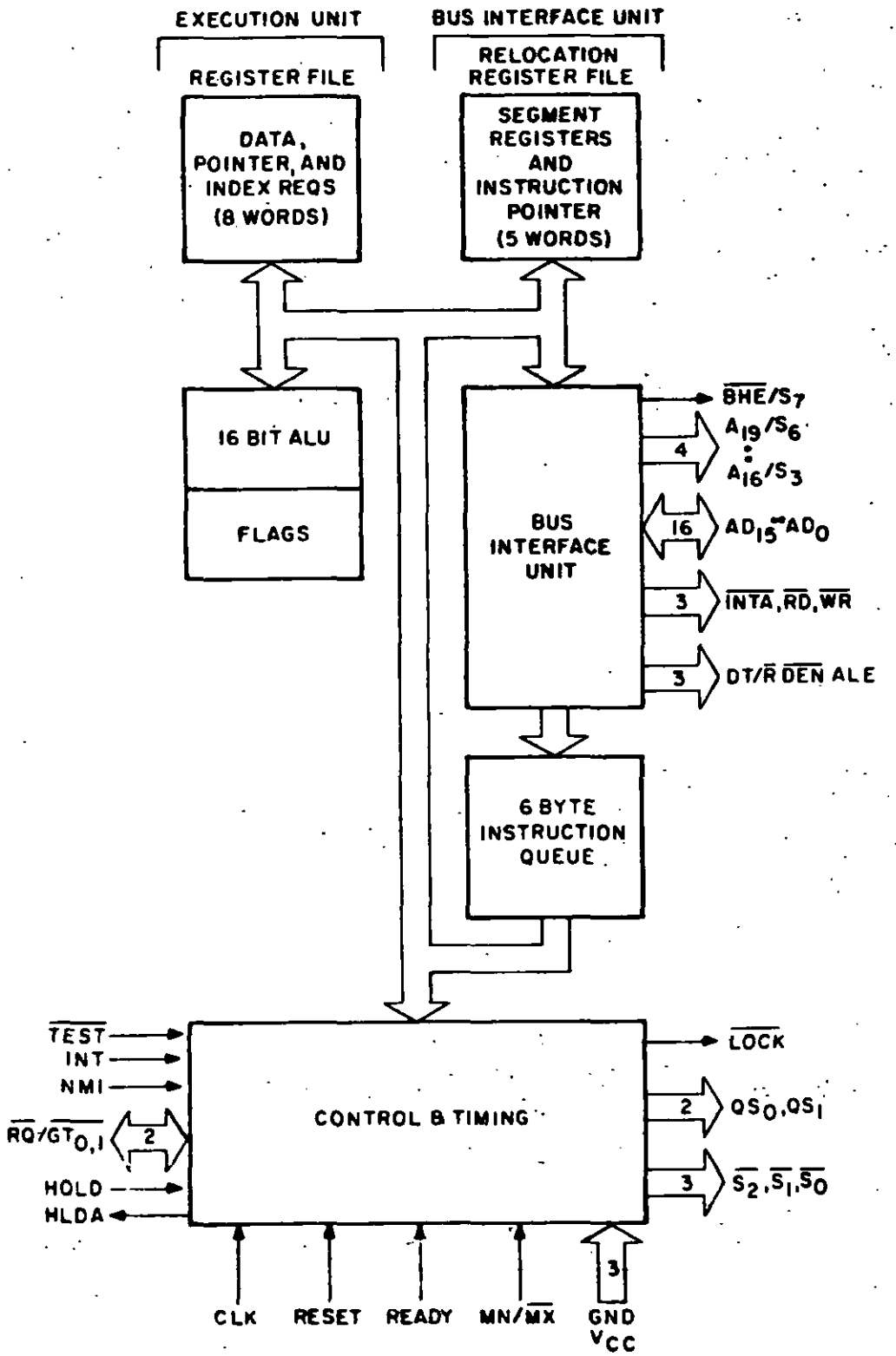
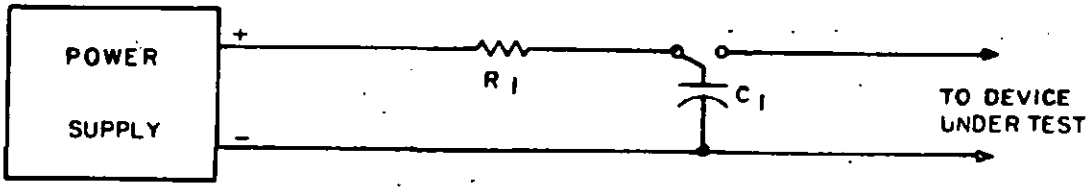


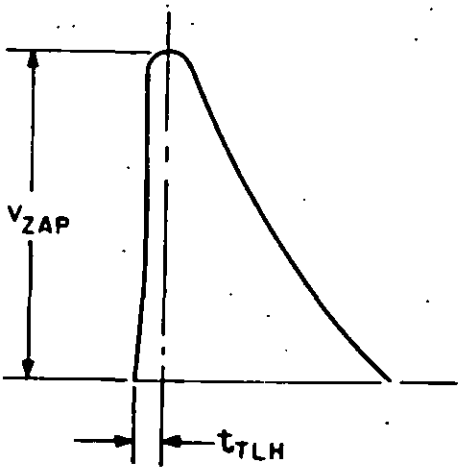

FIGURE 2. Functional block diagram.



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R1 = Appropriate current limiting resistance.
C1 = 100 pF ±20%.
$V_{zap}$ = 400 V measured at device input terminal.

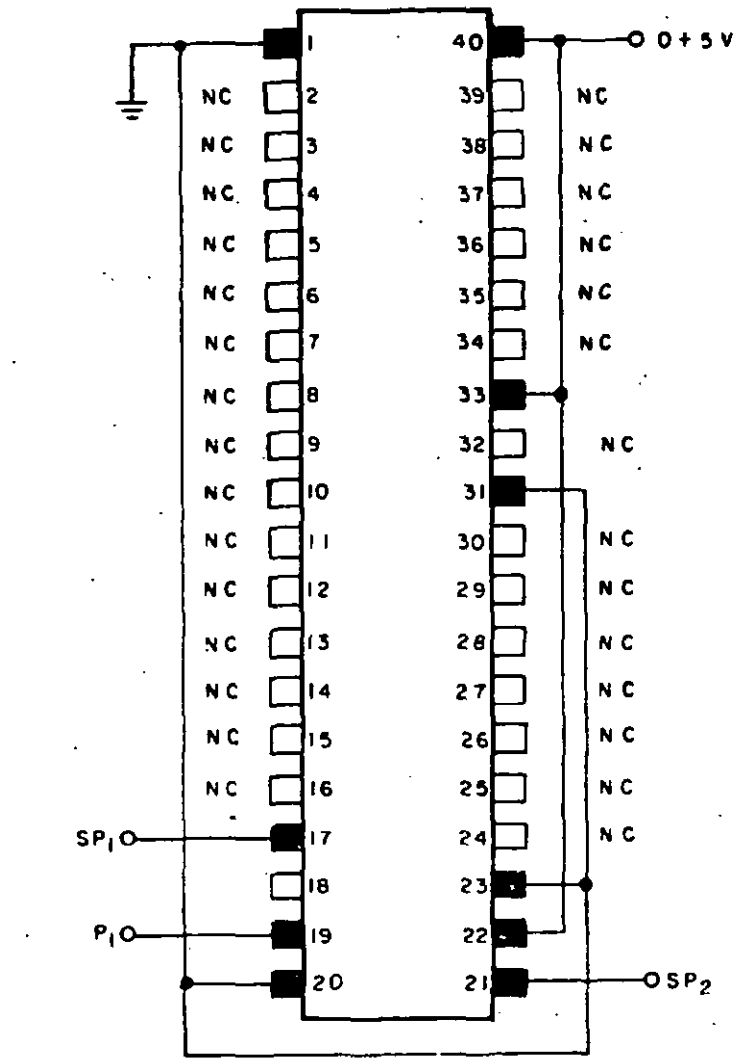



Pulse transition time ($t_{TLH}$) ≤ 50 ns (10%-90%).

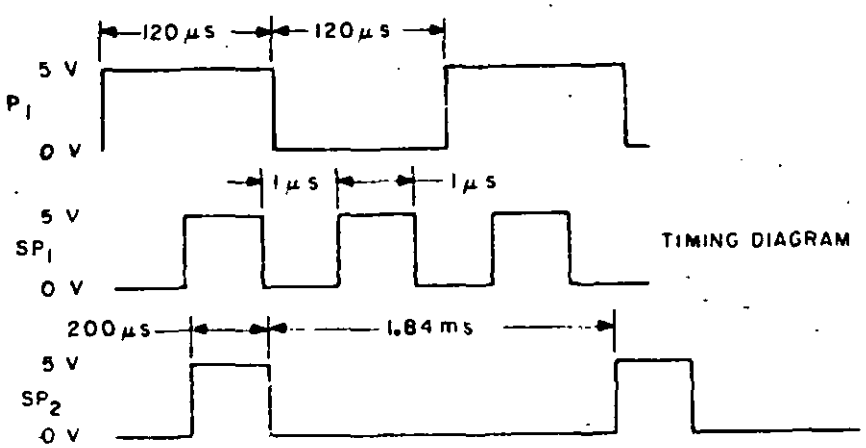
FIGURE 3. High voltage ($V_{zap}$) test circuits.



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| Pin | Pin name | Stress |
|---|---|---|
| 1 | $V_{SS}$ | GND |
| 2 | $AD_{14}$ | NC |
| 3 | $AD_{13}$ | NC |
| 4 | $AD_{12}$ | NC |
| 5 | $AD_{11}$ | NC |
| 6 | $AD_{10}$ | NC |
| 7 | $AD_9$ | NC |
| 8 | $AD_8$ | NC |
| 9 | $AD_7$ | NC |
| 10 | $AD_6$ | NC |
| 11 | $AD_5$ | NC |
| 12 | $AD_4$ | NC |
| 13 | $AD_3$ | NC |
| 14 | $AD_2$ | NC |
| 15 | $AD_1$ | NC |
| 16 | $AD_0$ | NC |
| 17 | NMI | SP1 |
| 18 | INTR | NC |
| 19 | CLK | P1 |
| 20 | $V_{SS}$ | GND |
| 21 | RESET | SP2 |
| 22 | READY | +5 V |
| 23 | $\overline{TEST}$ | GND |
| 24 | $\overline{INTA}$ | NC |
| 25 | ALE | NC |
| 26 | $\overline{DEN}$ | NC |
| 27 | DT/R | NC |
| 28 | M/IO | NC |
| 29 | $\overline{WR}$ | NC |
| 30 | HLDA | NC |
| 31 | HOLD | GND |
| 32 | $\overline{RD}$ | NC |
| 33 | MN/MX | +5 V |
| 34 | $\overline{BHE}/S_7$ | NC |
| 35 | $A_{19}/S_6$ | NC |
| 36 | $A_{18}/S_5$ | NC |
| 37 | $A_{17}/S_4$ | NC |
| 38 | $A_{16}/S_3$ | NC |
| 39 | $AD_{15}$ | NC |
| 40 | $V_{CC}$ | +5 V |




| Supply | Voltage | Tolerances | Current device | Noise levels |
|---|---|---|---|---|
| $V_{CC}$ | +5 V | ±.25 V | 150 mA | .4 V |

FIGURE 4. Burn-in and life test configuration.



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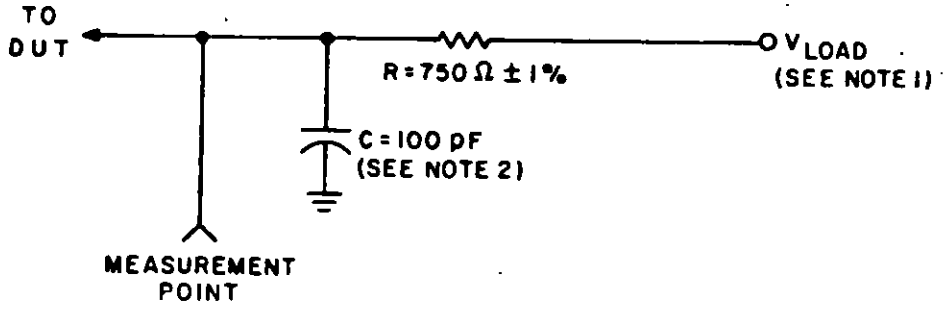
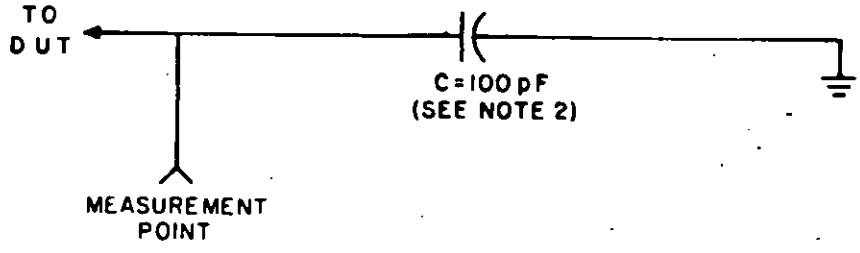



FIGURE 5. D.C. I/O load.

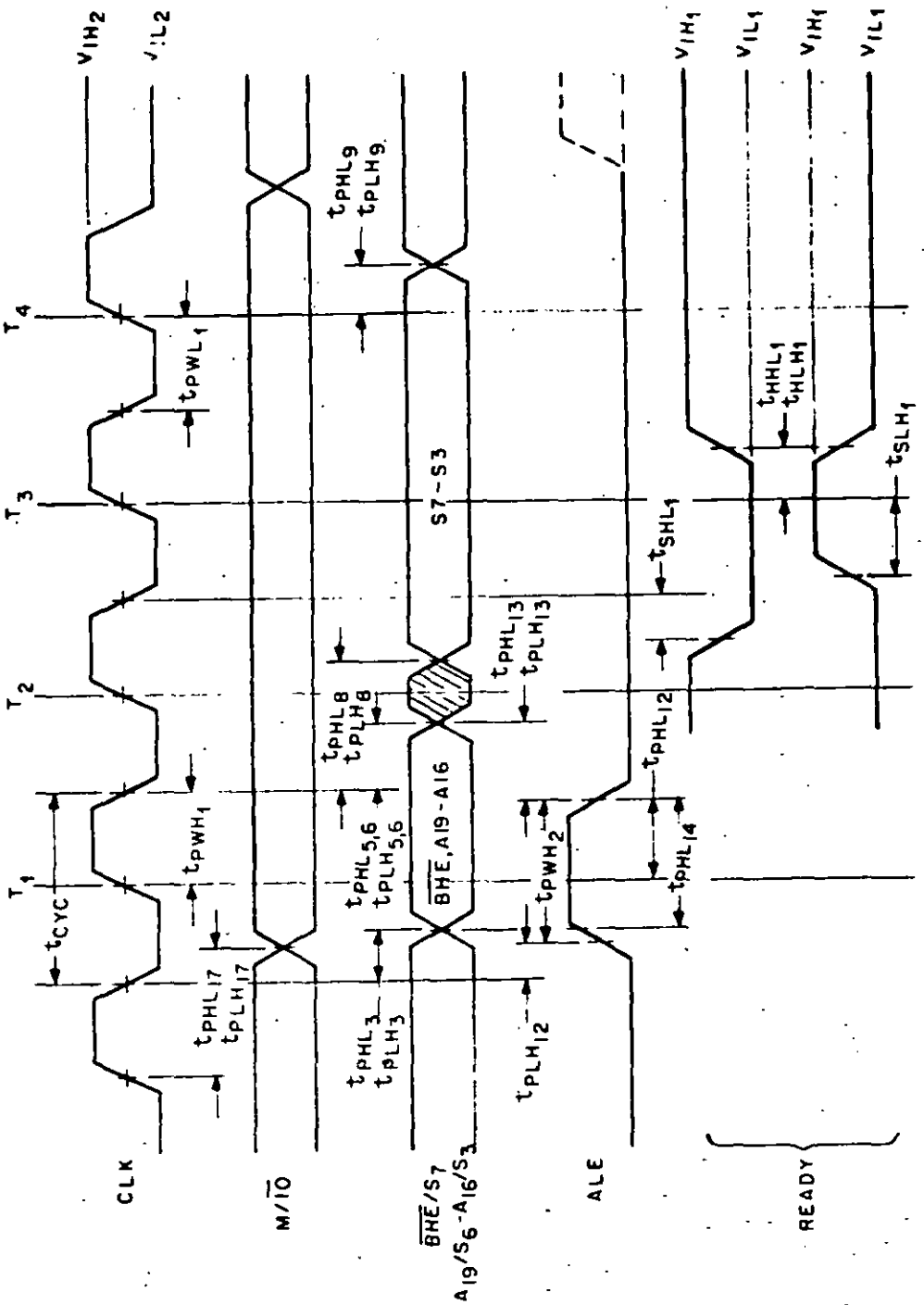



NOTES:
1. Loading used only during $V_{OL}$, $V_{OH}$ tests. $V_{LOAD}$ = 1.5 V for $V_{OL}$ test and $V_{LOAD}$ = 0.3 V for $V_{OH}$ test.
2. C includes tester, wiring and stray capacitance.

FIGURE 6. A.C. load.



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Check the source to verify that this is the current version before use.

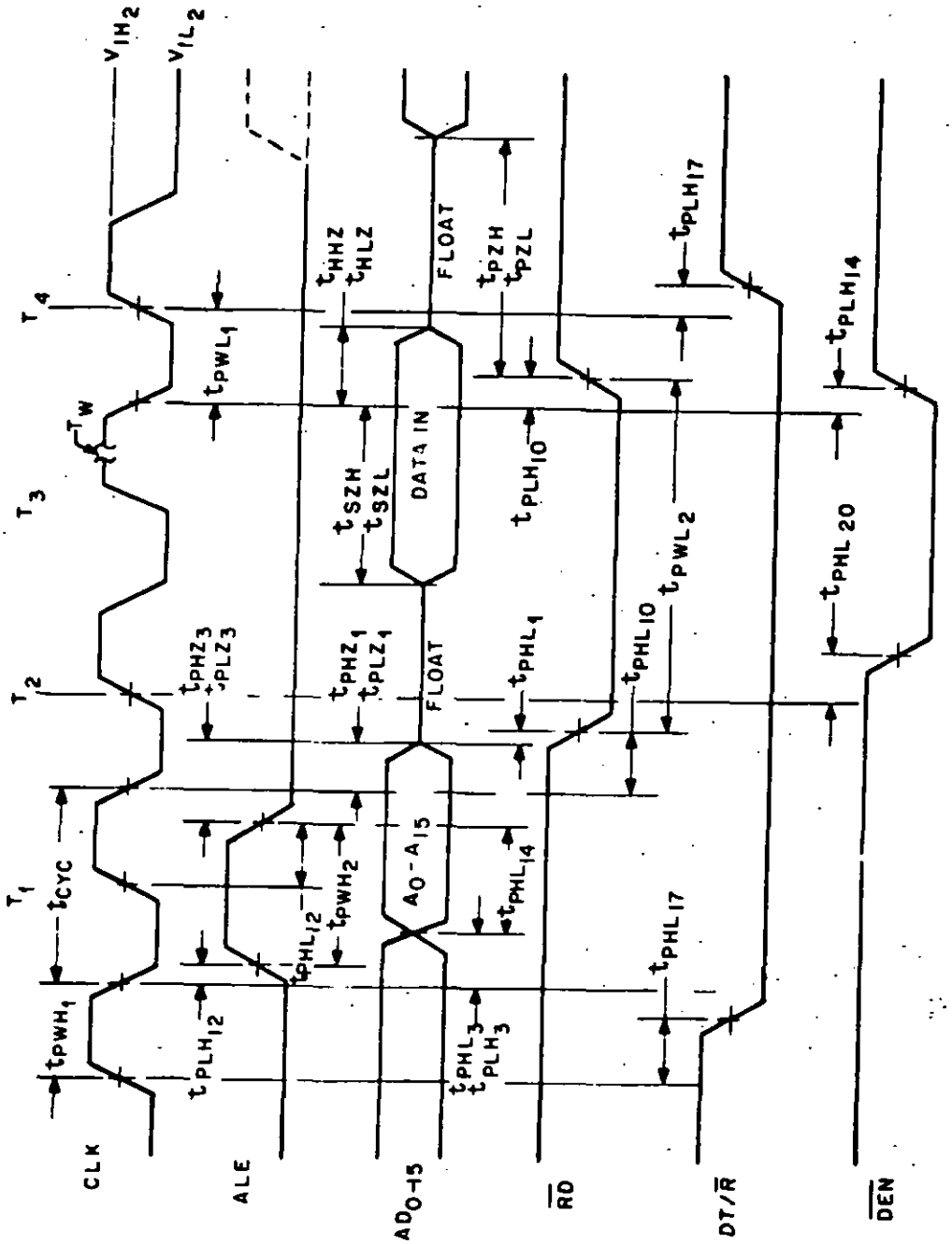



NOTE: All signals in figures 7, 8, 9, 10, 11, 12, 13, 14, and 15 switch between $V_{OL}$ and $V_{OH}$ unless otherwise specified.

FIGURE 7. Bus timing - minimum mode system.



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Check the source to verify that this is the current version before use.

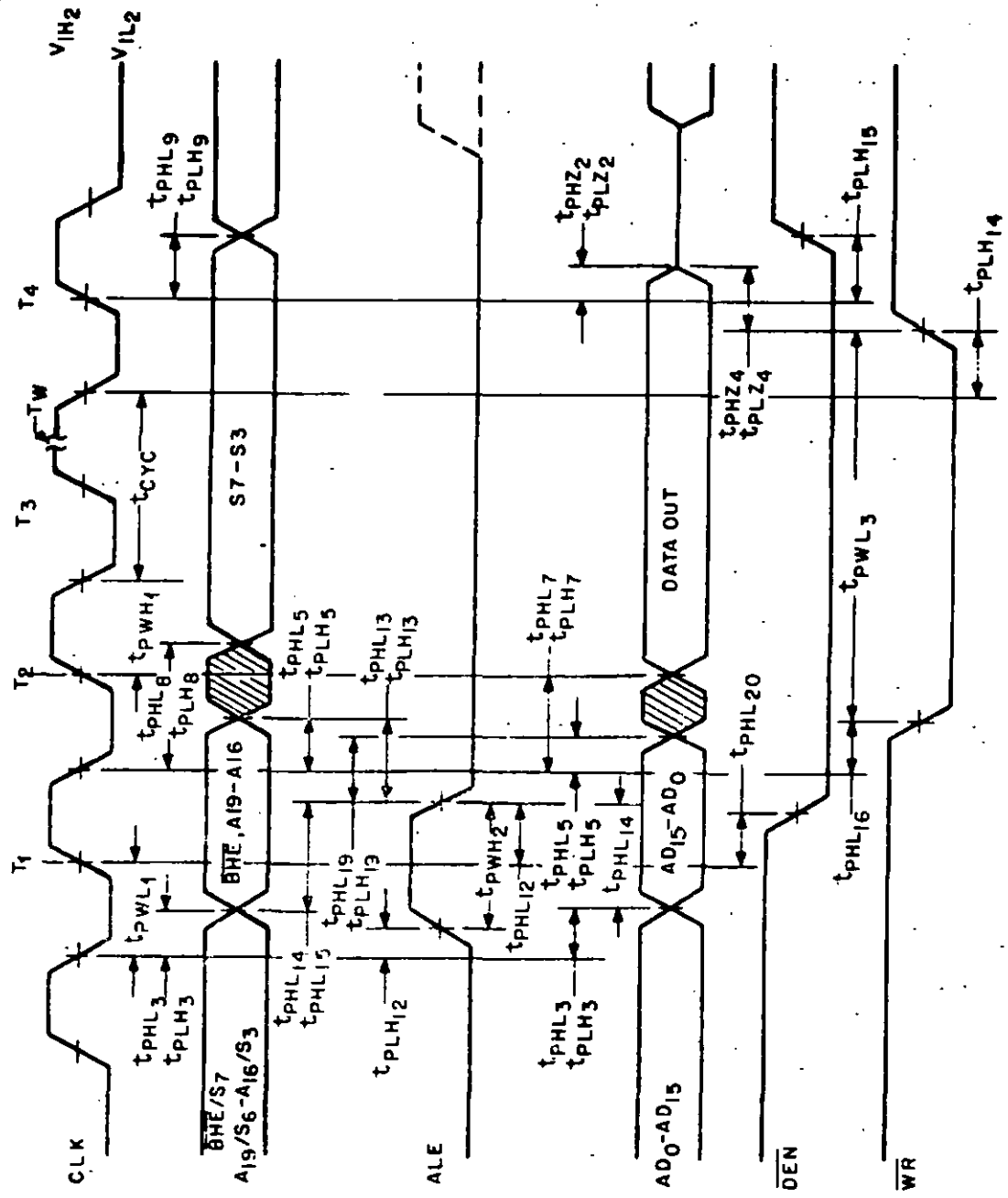



NOTE: $\overline{WR}$, $\overline{INTA}$ = $V_{OH}$.

FIGURE 8. Bus timing - minimum mode - read cycle.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:49Z
Check the source to verify that this is the current version before use.

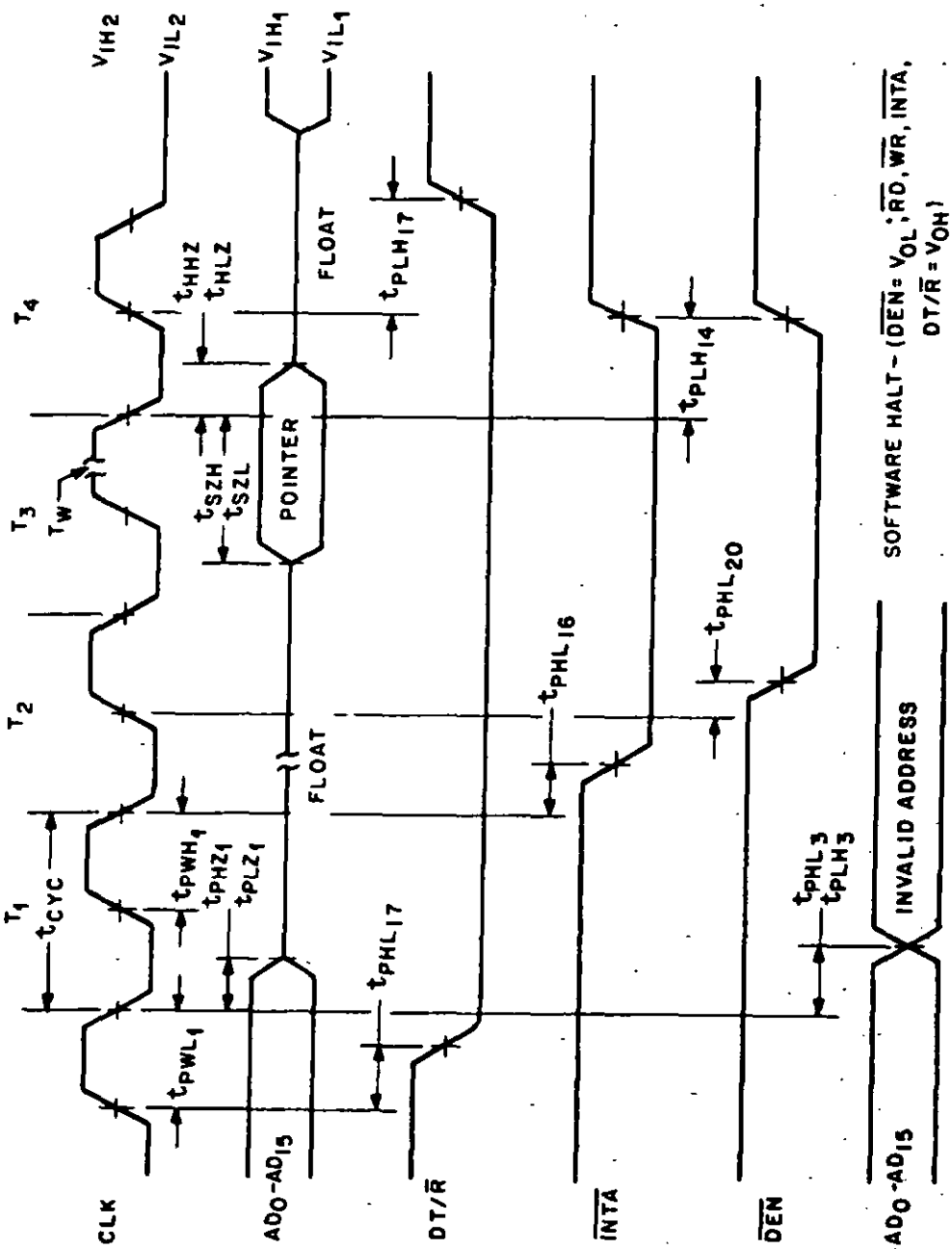



NOTE: $\overline{RD}$, $\overline{INTA}$ and $DT/\overline{R}$ = $V_{OH}$.

FIGURE 9. Bus timing - minimum mode system - write cycle.



Source: https://assist.dla.mil - Downloaded: 2026-10-21T14:49Z
Check the source to verify that this is the current version before use.

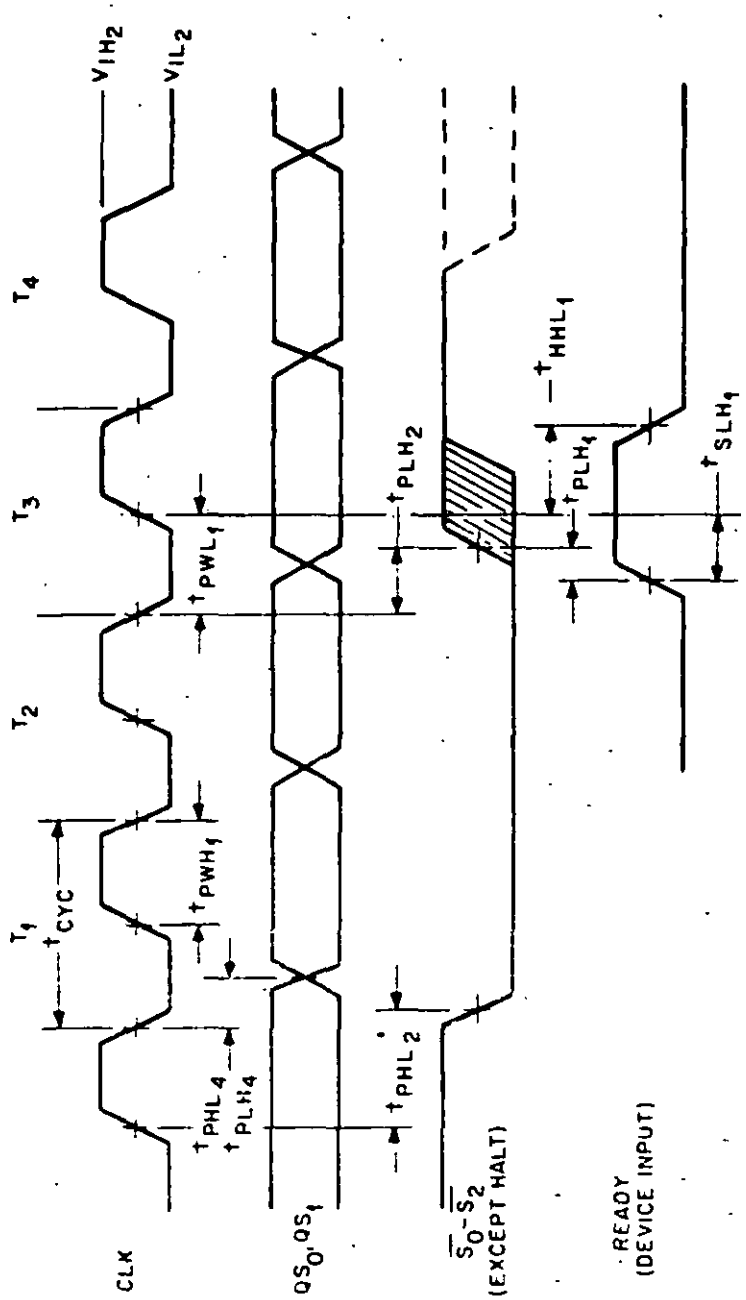



NOTES:
1. $\overline{RD}$, $\overline{WR}$ = $V_{OH}$; $\overline{BHE}$ = $V_{OL}$
2. Two INTA cycles run consecutively. The local ADDR/DATA bus is floating during both cycles. Control signals are shown for second INTA cycle.

FIGURE 10. Bus timing - minimum mode system - interrupt cycle.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:49Z
Check the source to verify that this is the current version before use.

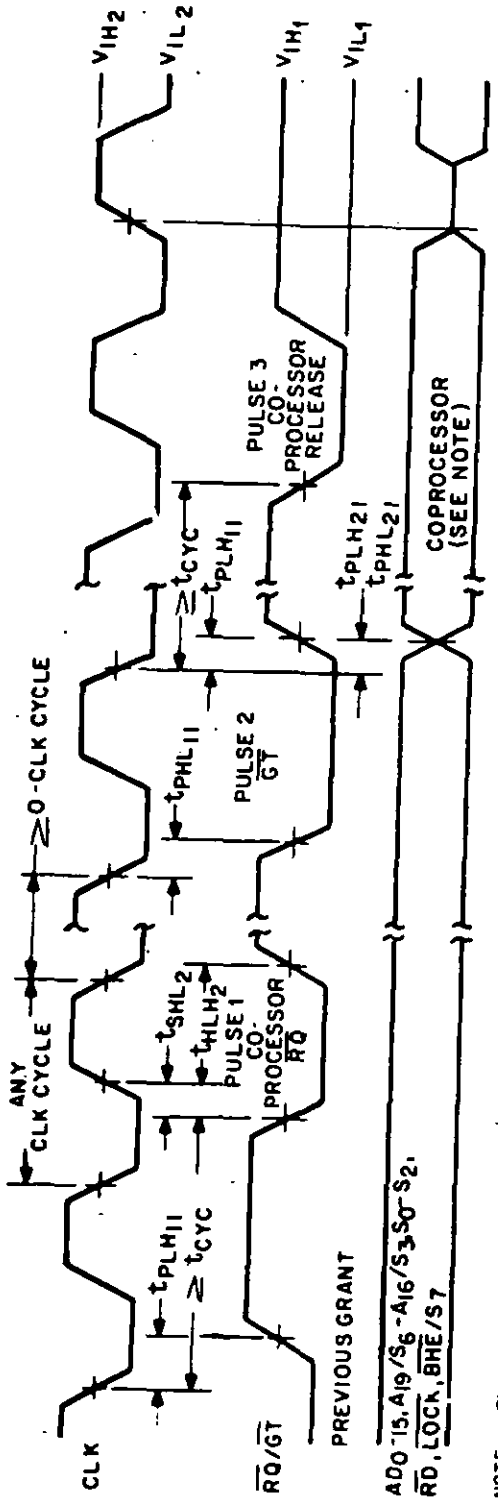



NOTE: Status inactive in state just prior to $T_4$.

FIGURE 11. Bus timing status signals (maximum mode).



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:49Z
Check the source to verify that this is the current version before use.




NOTE: The coprocessor may not drive the busses outside the region shown without risking contention.

FIGURE 12. Request/grant sequence timing (maximum mode only).

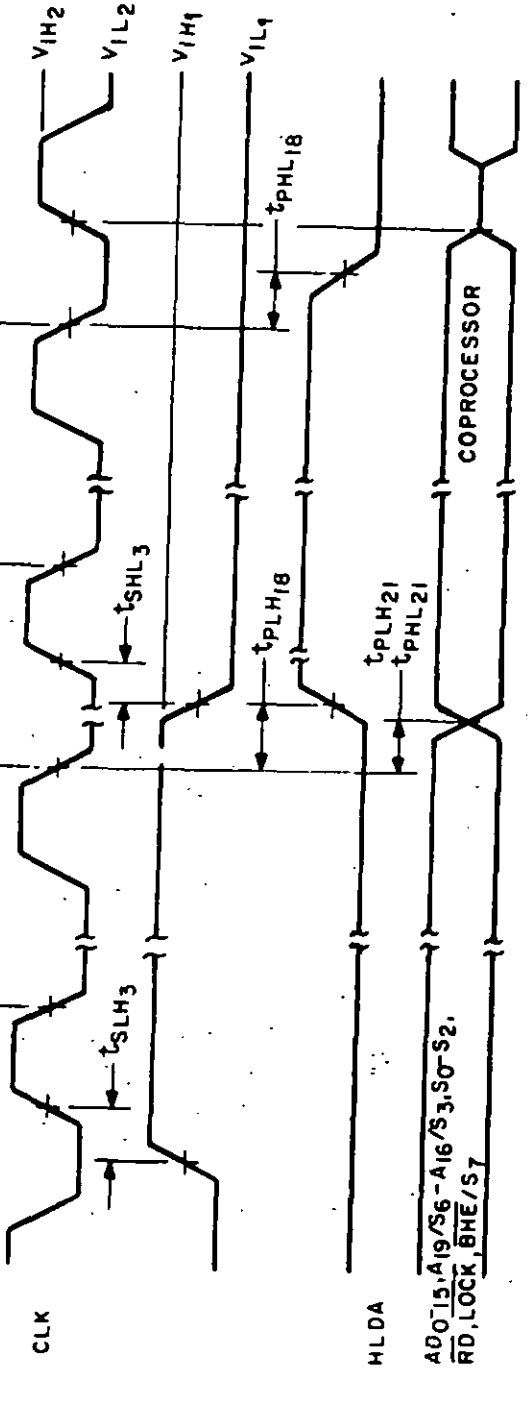
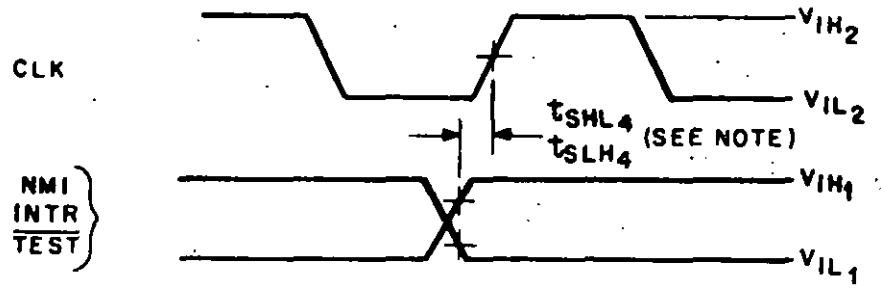



FIGURE 13. Hold/hold acknowledge timing (minimum mode only).



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Check the source to verify that this is the current version before use.




NOTE: Setup requirements for asynchronous signals only to ensure recognition at next clk.

FIGURE 14. Asynchronous signal recognition.

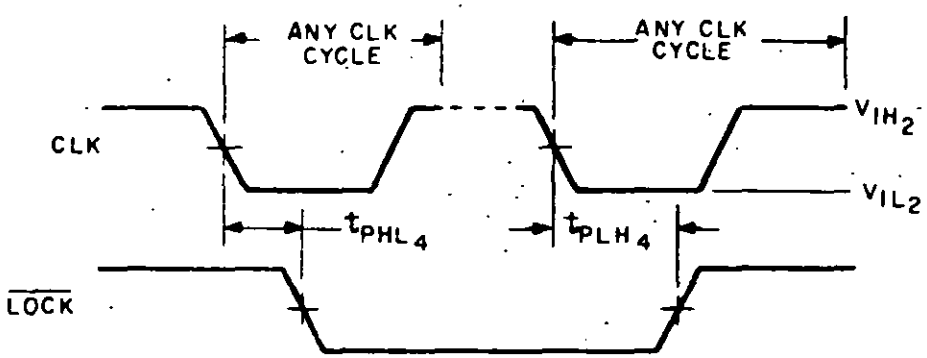



FIGURE 15. Bus lock signal timing (maximum mode only).



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TABLE IIIA. Input conditions for table IIIC.

| Symbol | Pin number | Terminal | Input conditions | | | | | | | | | Figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A | B | C | D | E | F | G | H | I | |
| $V_{IH1}$ | 2-16, 39 | All inputs except CLK | 2.4 | 2.2 | 3.0 | 2.4 | 2.4 | 3.0 | 3.0 | 3.0 | 3.0 | |
| $V_{IH2}$ | 19 | CLK | 4.0 | 4.0 | 4.0 | 3.9 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | |
| $V_{IL1}$ | 2-16, 39 | All inputs except CLK | 0.6 | 0.0 | 0.0 | .45 | .45 | 0.0 | 0.0 | 0.0 | 0.0 | |
| $V_{IL2}$ | 19 | CLK | 0.0 | 0.0 | 0.0 | 0.0 | 0.5 | 0.0 | 0.0 | 0.0 | 0.0 | |
| $t_{CYC}$ | 19 | CLK | 500 | 500 | 500 | 500 | 500 | 350 | 350 | 200 | 200 | 7,8,9, 10,11 |
| $t_{SZH}$ $t_{SZL}$ | 2-16, 39 | AD0-AD15 | 100 | 100 | 100 | 100 | 100 | 40 | 40 | 30 | 30 | 8,10 |
| $t_{HLZ}$ $t_{HHZ}$ | 2-16, 39 | AD0-AD15 | 100 | 100 | 100 | 100 | 100 | 30 | 30 | 10 | 10 | 8,10 |
| $t_{SLH4}$ $t_{SHL4}$ | 17, 18,23 | NMI INTR TEST | 100 | 100 | 100 | 100 | 100 | 40 | 40 | 30 | 30 | 14 |
| $t_{SLH1}$ | 22 | RDY | 395 | 395 | 395 | 395 | 395 | 238 | 251 | 108 | 121 | 7,11 |
| $t_{HLH1}$ $t_{HHL1}$ | 22 | RDY | 105 | 105 | 105 | 105 | 105 | 112 | 99 | 92 | 79 | 7,11 |
| $t_{SHL1}$ | 22 | RDY | 50 | 50 | 50 | 50 | 50 | 320 | 30 | 190 | 190 | 7 |
| $t_{SHL2}$ | 31 | $\overline{RQ}/\overline{GT0}$ | 155 | 155 | 155 | 155 | 155 | 40 | 40 | 30 | 30 | 12 |
| $t_{HLH2}$ | 31 | $\overline{RQ}/\overline{GT0}$ | 345 | 345 | 345 | 345 | 345 | 310 | 241 | 170 | 170 | 12 |
| $t_{SLH3}$ $t_{SHL3}$ | 30, 31 | $\overline{RQ}/\overline{GT0}$ $\overline{RQ}/\overline{GQ1}$ | 100 | 100 | 100 | 100 | 100 | 40 | 40 | 30 | 30 | 13 |
| $t_{PWH1}$ | 19 | CLK | 155 | 155 | 155 | 155 | 155 | 82 | 67 | 82 | 52 | 7,8,9, 10,11 |



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TABLE IIIB. Output conditions for table IIIC.

| Symbol | Waveform figure | Terminal | Output conditions | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | A | B | C | D | E | F | |
| $V_{OH}$ | | | 2.0 | 2.4 | 1.5 | 1.5 | 1.5 | 1.5 | V |
| $V_{OL}$ | | | 1.0 | 0.45 | 1.5 | 1.5 | 1.5 | 1.5 | V |
| $t_{PHL7}$ $t_{PLH7}$ | 9 | AD0-15 | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL8}$ $t_{PLH8}$ | 7,9 | $\overline{BHE}$/S7, A16/S3-A19/S6 | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL10}$ $t_{PLH10}$ | 8 | $\overline{RD}$ | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL11}$ $t_{PLH11}$ | 12 | $\overline{RQ}/\overline{GT0}$, I | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL12}$ | 7,8,9 | ALE | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| $t_{PLH24}$ | 9 | $\overline{DEN}$ | 460 | 460 | 200 | 200 | 200 | 200 | ns |
| $t_{PHL16}$ | 9,10 | $\overline{WR}$ | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| | | $\overline{INTA}$ | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| $t_{PHL17}$ $t_{PLH17}$ | 7,8,9 | $\overline{DT}/\overline{R}$, M/$\overline{IO}$ | 460 | 460 | 200 | 200 | 200 | 200 | ns |
| $t_{PHL18}$ $t_{PLH18}$ | 13 | HLDA | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL2}$ | 11 | $\overline{S2}$-$\overline{S0}$ | 460 | 460 | 200 | 200 | 200 | 200 | ns |
| $t_{PLH2}$ | 11 | $\overline{S2}$-$\overline{S0}$ | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| $t_{PLH3}$ $t_{PHL3}$ | 7,8,9, 10 | AD0-AD15, A16/S3-A19/S6 | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PLH4}$ | 11,15 | QS1, QS0, LOCK | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| $t_{PHL4}$ | | $\overline{BHE}$/S7 | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PHL12}$ | 7,8,9 | ALE | 460 | 460 | 200 | 200 | 200 | 200 | ns |
| $t_{PHL20}$ | 8,9.10 | $\overline{DEN}$ | 460 | 460 | 200 | 200 | 200 | 200 | ns |

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TABLE IIIB. Output conditions for table IIIC - Continued.

| Symbol | Waveform figure | Terminal | Output conditions | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | A | B | C | D | E | F | |
| $t_{PHL21}$ | 12,13 | $\overline{S2}$-$\overline{S0}$, | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| $t_{PLH21}$ | | AD0-AD15, A16/S3-A19/S6, $\overline{RD}$, $\overline{LOCK}$, $\overline{BHE}$/S7 | 460 | 460 | 140 | 140 | 120 | 120 | ns |
| $t_{PLH23}$ | 8,9,10 | $\overline{DEN}$, $\overline{INTA}$ | 305 | 305 | 118 | 131 | 118 | 131 | ns |
| | | $\overline{WR}$ | 460 | 460 | 140 | 140 | 120 | 120 | ns |



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4.5.3 High-voltage ($V_{ZAP}$) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 V from a 100 pF source in the following test sequence.

a. Measure $I_{IH}$ and $I_{IL}$ at one input terminal of the device under tests at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of $I_{IH}$ and $I_{IL}$ shall be as specified in table III.

b. Apply the test voltage ($V_{ZAP}$ = 400 V) to the same terminal selected for leakage current measurements. Apply $V_{ZAP}$ in a 2-pulse sequence as follows and in accordance with the test circuit of figure 3.

1. Input (-) to $V_{SS}$.

2. Input (+) to $V_{SS}$.

c. Within 24 hours repeat the $I_{IH}$ and $I_{IL}$ measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.

4.6 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Test vector table. Table IV, which forms a part of this specification, is not printed herein because of its extreme length and complexity; and, even when presented as hard copy, it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture (magnetic tape format usually preferred). The preparing activity (or DESC, as its agent) may be consulted for information on obtaining such output. For referee purposes, Rome Air Development Center (RADC-RBRA) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited quantity of dated, hard copy printouts of this table.

6.3 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.4 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the data (see 4.6) pertinent to the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to procuring activity in addition to notification to qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicble.

f. Requirements for product assurance options.

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TABLE IIIC. Group A
Terminal conditions (pins not

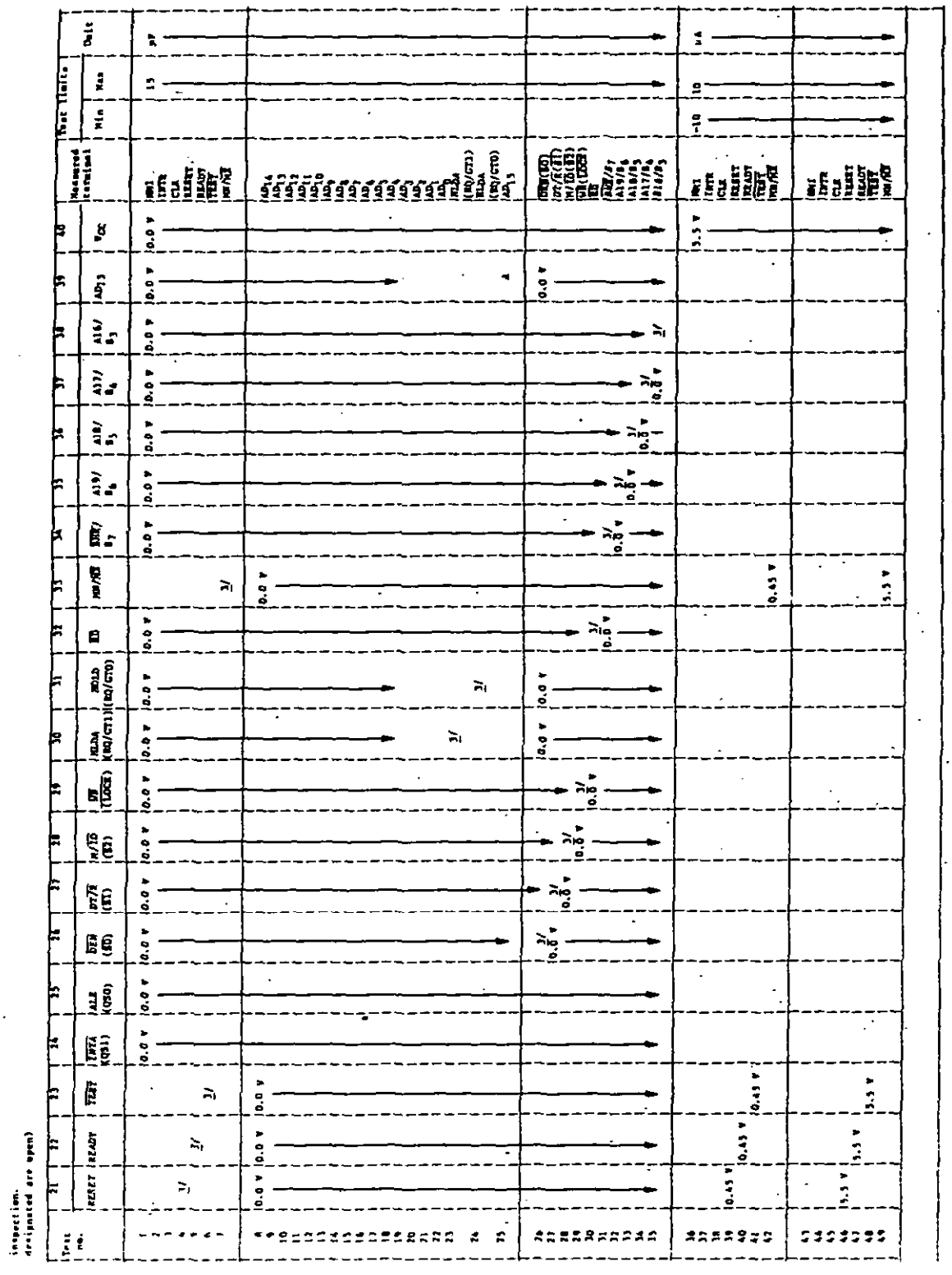
| Subgroup | Symbol | MIL-STD-883 method | Test no. | 1 CMD | 2 $AD_{14}$ | 3 $AD_{13}$ | 4 $AD_{12}$ | 5 $AD_{11}$ | 6 $AD_{10}$ | 7 $AD_9$ | 8 $AD_8$ | 9 $AD_7$ | 10 $AD_6$ | 11 $AD_5$ | 12 $AD_4$ | 13 $AD_3$ | 14 $AD_2$ | 15 $AD_1$ | 16 $AD_0$ | 17 $\overline{BE1}$ | 18 $\overline{BE0}$ | 19 CLK | 20 CMD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 1/ | 0.0 V |
| | | | 2 | | | | | | | | | | | | | | | | | | | | |
| | | | 3 | | | | | | | | | | | | | | | | | | | | |
| | | | 4 | | | | | | | | | | | | | | | | | | | | |
| | | | 5 | | | | | | | | | | | | | | | | | | | | |
| | | | 6 | | | | | | | | | | | | | | | | | | | | |
| | | | 7 | | | | | | | | | | | | | | | | | | | | |
| | | | 8 | | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 2/ | 0.0 V | 0.0 V | 0.0 V | |
| | | | 9 | | | | | | | | | | | | | | | | | | | | |
| | | | 10 | | | | | | | | | | | | | | | | | | | | |
| | | | 11 | | | | | | | | | | | | | | | | | | | | |
| | | | 12 | | | | | | | | | | | | | | | | | | | | |
| | | | 13 | | | | | | | | | | | | | | | | | | | | |
| | | | 14 | | | | | | | | | | | | | | | | | | | | |
| | | | 15 | | | | | | | | | | | | | | | | | | | | |
| | | | 16 | | | | | | | | | | | | | | | | | | | | |
| | | | 17 | | | | | | | | | | | | | | | | | | | | |
| | | | 18 | | | | | | | | | | | | | | | | | | | | |
| | | | 19 | | | | | | | | | | | | | | | | | | | | |
| | | | 20 | | | | | | | | | | | | | | | | | | | | |
| | | | 21 | | | | | | | | | | | | | | | | | | | | |
| | | | 22 | | | | | | | | | | | | | | | | | | | | |
| | | | 23 | | | | | | | | | | | | | | | | | | | | |
| | | | 24 | | | | | | | | | | | | | | | | | | | | |
| | | | 25 | | | | | | | | | | | | | | | | | | | | |
| | | | 26 | | | | | | | | | | | | | | | | | | | | |
| | | | 27 | | | | | | | | | | | | | | | | | | | | |
| | | | 28 | | | | | | | | | | | | | | | | | | | | |
| | | | 29 | | | | | | | | | | | | | | | | | | | | |
| | | | 30 | | | | | | | | | | | | | | | | | | | | |
| | | | 31 | | | | | | | | | | | | | | | | | | | | |
| | | | 32 | | | | | | | | | | | | | | | | | | | | |
| | | | 33 | | | | | | | | | | | | | | | | | | | | |
| | | | 34 | | | | | | | | | | | | | | | | | | | | |
| | | | 35 | | | | | | | | | | | | | | | | | | | | |
| | | | 36 | | | | | | | | | | | | | | | | | 0.45 V | 0.45 V | 0.65 V | |
| | | | 37 | | | | | | | | | | | | | | | | | | | | |
| | | | 38 | | | | | | | | | | | | | | | | | | | | |
| | | | 39 | | | | | | | | | | | | | | | | | | | | |
| | | | 40 | | | | | | | | | | | | | | | | | | | | |
| | | | 41 | | | | | | | | | | | | | | | | | | | | |
| | | | 42 | | | | | | | | | | | | | | | | | | | | |
| | | | 43 | | | | | | | | | | | | | | | | | 5.5 V | 5.5 V | 5.5 V | |
| | | | 44 | | | | | | | | | | | | | | | | | | | | |
| | | | 45 | | | | | | | | | | | | | | | | | | | | |
| | | | 46 | | | | | | | | | | | | | | | | | | | | |
| | | | 47 | | | | | | | | | | | | | | | | | | | | |
| | | | 48 | | | | | | | | | | | | | | | | | | | | |
| | | | 49 | | | | | | | | | | | | | | | | | | | | |

Test pattern 4, vectors 4493-4506 of table IV, stop at vector 4506 and with CLK still applied perform the following tests.

See footnotes at end of table.



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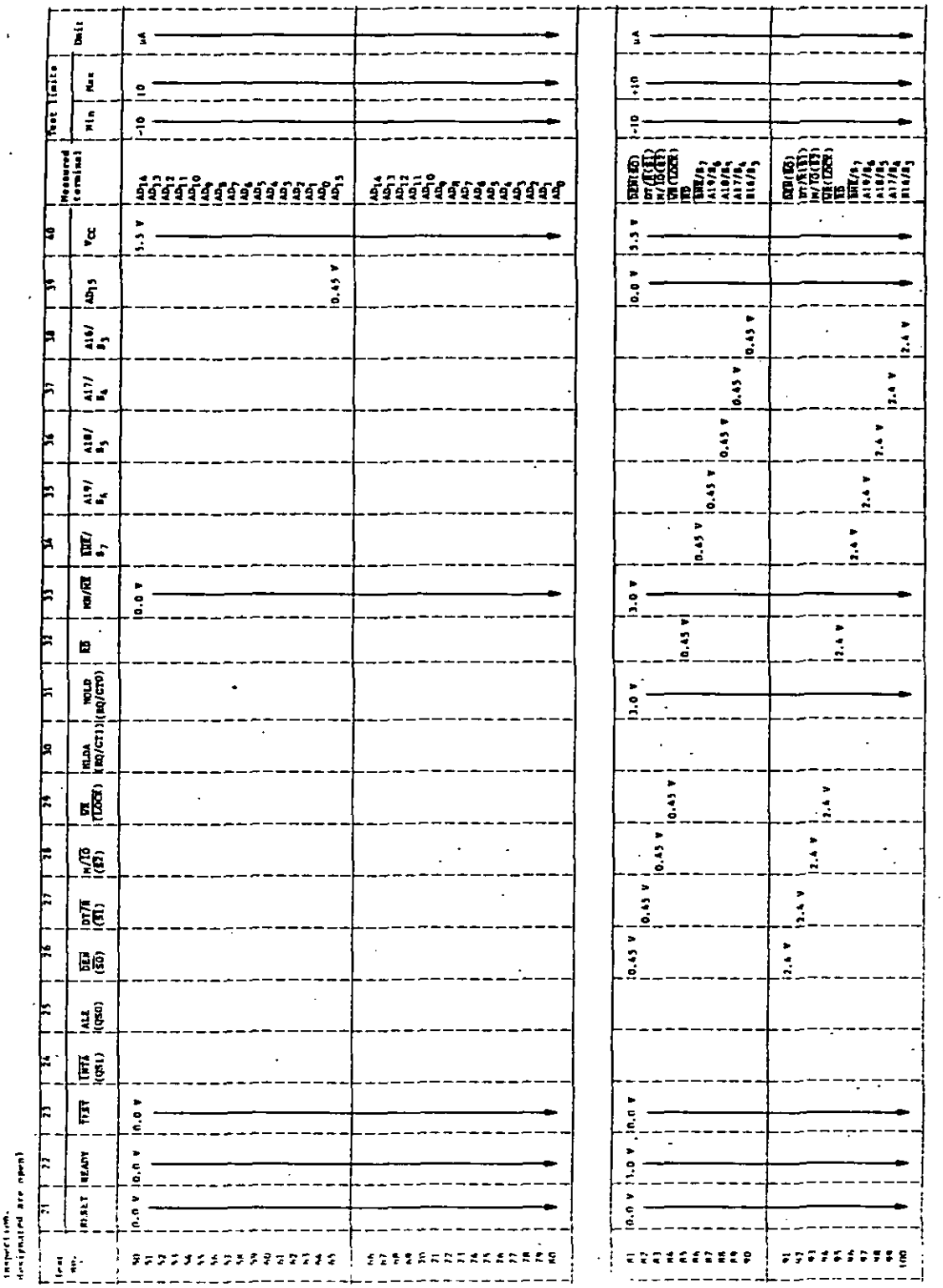
TABLE IIIC. Group A
Terminal conditions (pins not

| Subgroup | Symbol | MIL-STD-883 method | Test no. | 1 CSO | 2 $AD_{14}$ | 3 $AD_{13}$ | 4 $AD_{12}$ | 5 $AD_{11}$ | 6 $AD_{10}$ | 7 $AD_9$ | 8 $AD_8$ | 9 $AD_7$ | 10 $AD_6$ | 11 $AD_5$ | 12 $AD_4$ | 13 $AD_3$ | 14 $AD_2$ | 15 $AD_1$ | 16 $AD_0$ | 17 WE1 | 18 STB | 19 CLK | 20 GND |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Run patterns k, vectors 4945-4984 of table IV, stop at vector 4984 and with CLK still applied perform the following tests. | | | | | | | | | | | | | | | | | | |
| 1 | $I_{IL2}$ | | 50 | 10.0 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.45 V | 0.0 V | 0.0 V | 0/ | 0.0 V |
| | | | 51 | | | | | | | | | | | | | | | | | | | | |
| | | | 52 | | | | | | | | | | | | | | | | | | | | |
| | | | 53 | | | | | | | | | | | | | | | | | | | | |
| | | | 54 | | | | | | | | | | | | | | | | | | | | |
| | | | 55 | | | | | | | | | | | | | | | | | | | | |
| | | | 56 | | | | | | | | | | | | | | | | | | | | |
| | | | 57 | | | | | | | | | | | | | | | | | | | | |
| | | | 58 | | | | | | | | | | | | | | | | | | | | |
| | | | 59 | | | | | | | | | | | | | | | | | | | | |
| | | | 60 | | | | | | | | | | | | | | | | | | | | |
| | | | 61 | | | | | | | | | | | | | | | | | | | | |
| | | | 62 | | | | | | | | | | | | | | | | | | | | |
| | | | 63 | | | | | | | | | | | | | | | | | | | | |
| | | | 64 | | | | | | | | | | | | | | | | | | | | |
| | | | 65 | | | | | | | | | | | | | | | | | | | | |
| 1 | $I_{IH2}$ | | 66 | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | 15.5 V | | | | |
| | | | 67 | | | | | | | | | | | | | | | | | | | | |
| | | | 68 | | | | | | | | | | | | | | | | | | | | |
| | | | 69 | | | | | | | | | | | | | | | | | | | | |
| | | | 70 | | | | | | | | | | | | | | | | | | | | |
| | | | 71 | | | | | | | | | | | | | | | | | | | | |
| | | | 72 | | | | | | | | | | | | | | | | | | | | |
| | | | 73 | | | | | | | | | | | | | | | | | | | | |
| | | | 74 | | | | | | | | | | | | | | | | | | | | |
| | | | 75 | | | | | | | | | | | | | | | | | | | | |
| | | | 76 | | | | | | | | | | | | | | | | | | | | |
| | | | 77 | | | | | | | | | | | | | | | | | | | | |
| | | | 78 | | | | | | | | | | | | | | | | | | | | |
| | | | 79 | | | | | | | | | | | | | | | | | | | | |
| | | | 80 | | | | | | | | | | | | | | | | | | | | |
| | | | | Run pattern l, vectors 1-1484 of table IV, stop at vector 1484 and with CLK still applied perform the following tests. | | | | | | | | | | | | | | | | | | |
| 1 | $I_{ILI}$ | | 81 | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0.0 V | 0/ | 0.0 V |
| | | | 82 | | | | | | | | | | | | | | | | | | | | |
| | | | 83 | | | | | | | | | | | | | | | | | | | | |
| | | | 84 | | | | | | | | | | | | | | | | | | | | |
| | | | 85 | | | | | | | | | | | | | | | | | | | | |
| | | | 86 | | | | | | | | | | | | | | | | | | | | |
| | | | 87 | | | | | | | | | | | | | | | | | | | | |
| | | | 88 | | | | | | | | | | | | | | | | | | | | |
| | | | 89 | | | | | | | | | | | | | | | | | | | | |
| | | | 90 | | | | | | | | | | | | | | | | | | | | |
| 1 | $I_{IHI}$ | | 91 | | | | | | | | | | | | | | | | | | | | |
| | | | 92 | | | | | | | | | | | | | | | | | | | | |
| | | | 93 | | | | | | | | | | | | | | | | | | | | |
| | | | 94 | | | | | | | | | | | | | | | | | | | | |
| | | | 95 | | | | | | | | | | | | | | | | | | | | |
| | | | 96 | | | | | | | | | | | | | | | | | | | | |
| | | | 97 | | | | | | | | | | | | | | | | | | | | |
| | | | 98 | | | | | | | | | | | | | | | | | | | | |
| | | | 99 | | | | | | | | | | | | | | | | | | | | |
| | | | 100 | | | | | | | | | | | | | | | | | | | | |

See footnotes at end of table.



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TABLE IIIC. Group A
Terminal conditions (pins not

| Subgroup | Symbol | Parameter verification | Test no. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | CP2 | $A^{*}14$ | $AD_{13}$ | $AD_{12}$ | $AD_{11}$ | $AD_{10}$ | $AD_9$ | $AD_8$ | $AD_7$ | $AD_6$ | $AD_5$ | $AD_4$ | $AD_3$ | $AD_2$ | $AD_1$ | $AD_0$ | HRQ | IPT0 | CLK | GND |
| 1<br>$T_C$ = 25°C | $I_{CC}$ | | 101 | Run pattern 3 of table IV using input condition C of table IIIA and $V_{CC}$ = 5.5 V. Monitor $I_{CC}$ throughout pattern to verify that it does not exceed limit, using | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | | 102 | Run pattern 2 of table IV using input condition C of table IIIA and $V_{CC}$ = 5.5 V, and output condition B of table IIIB, with $V_{LOAD}$ = 1.95 V (see figure 5) | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | | 103 | Run pattern 2 of table IV using input condition C of table IIIA and $V_{CC}$ = 4.5 V, and output condition B of table IIIB, with $V_{LOAD}$ = 2.1 V (see figure 5) | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | | 104 | Run pattern 1 of table IV using input condition A of table IIIA and $V_{CC}$ = 4.5 V, and output condition A of table IIIB. | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | $V_{IL1}$ | 105 | Run pattern 1 of table IV using input condition B of table IIIA and $V_{CC}$ = 5.5 V, and output condition A of table IIIB. | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | $V_{IL2}$ | 106 | Run pattern 1 of table IV using input condition B of table IIIA and $V_{CC}$ = 4.5 V, and output condition A of table IIIB. | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | $V_{IH1}$ | 107 | Run pattern 1 of table IV using input condition B of table IIIA and $V_{CC}$ = 5.5 V, and output condition A of table IIIB. | | | | | | | | | | | | | | | | | | | |
| | $V_{OL}$<br>$V_{OH}$ | $V_{IH2}$ | | | | | | | | | | | | | | | | | | | | | |
| 2 | Repeat subgroup 1, cases 26-77 at $T_C$ = 125°C, except limit for test 71 shall be as shown in limit column | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Repeat subgroup 1, cases 26-77 at $T_C$ = -55°C, except limit for test 71 shall be as shown in limit column | | | | | | | | | | | | | | | | | | | | | | |
| 7 | Run pattern 8 of table IV using input condition C of table IIIA and $V_{CC}$ = 4.5 V and 5.5 V, and output condition A of table IIIB. | | | | | | | | | | | | | | | | | | | | | | |
| 8 | Repeat subgroup 7 at -55°C and +125°C. | | | | | | | | | | | | | | | | | | | | | | |

See footnote at end of table.



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| Test no. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RESET | WAIT | WRITE | $I_{OZH}$ (CS1) | $I_{OZL}$ (CS0) | $I_{OZR}$ (RD) | $I_{OZW}$ (WR) | $I_{OZ}$ (DBIN) | CK (CLOCK) | $I_{IL}$ (SYNC) | | Min | Max | |
| 101 | Output condition A of table III. | | | | | | | | | | | 340 | | mA |
| 102 | | | | | | | | | | | | 2.22 | 0.45 | V |
| 103 | | | | | | | | | | | | 2.4 | 0.6 | V |
| 104 | | | | | | | | | | | | 2.4 | 0.45 | V |
| 105 | | | | | | | | | | | | 2.4 | 0.45 | V |
| 106 | | | | | | | | | | | | 2.4 | 0.45 | V |
| 107 | | | | | | | | | | | | 2.4 | 0.45 | V |
| | | | | | | | | | | | | 190 | | mA |
| | | | | | | | | | | | | 433 | | mA |

Inputs not designated are open)



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TABLE III C. Group A inspection - Continued.

| Symbol | Test number | Pattern number | Waveform figure | Parameter verified | Input cond. from table III A | Output cond. from table III B | Measurement terminal: Input under test | Measurement terminal: Output under test | Test limits [1/]: Subgroup 9 $T_A$ = 25°C: Min | Subgroup 9 $T_A$ = 25°C: Max | Subgroup 10 $T_C$ = 125°C: Min | Subgroup 10 $T_C$ = 125°C: Max | Subgroup 11 $T_C$ = -55°C: Min | Subgroup 11 $T_C$ = -55°C: Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $t_{PLH1}$ | 108 | | 11 | — | | — | Ready | $A16/S_3$-$\overline{MEX}/S_7$ | — | 110 | — | 110 | — | 110 | ns |
| $t_{PHL2}$ | 109 [4/] | 14 | 11 | — | F | — | CLK | $S_0$-$S_2$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PLH2}$ | 110 [4/] | 26 | 11 | — | G | — | CLK | $S_0$-$S_2$ | 10 | 130 | 10 | 130 | 10 | 130 | ns |
| $t_{PHL3}$ $t_{PLH3}$ | 111 [4/] | 38 | 7,8,9,10 | — | G | — | CLK | $AD_0$-$AD_{15}$ $A16/S_3$-$A19/S_6$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL4}$ $t_{PLH4}$ | 112 [4/] | 28, 29, 40 | 11,15 | — | G | — | CLK | $\overline{QS0}$,$\overline{QS1}$ $\overline{LOCK}$, $\overline{MEX}/S_7$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL5}$ $t_{PLH5}$ | 113 | | 9,7 | — | | — | CLK | $AD_0$-$AD_{15}$ $A16/S_3$-$A19/S_6$ | 10 | — | 10 | — | 10 | — | ns |
| $t_{PHL6}$ $t_{PLH6}$ | 114 | | 7 | — | | — | CLK | $\overline{MEX}/S_7$ | 10 | — | 10 | — | 10 | — | ns |
| $t_{PHL7}$ $t_{PLH7}$ | 115 [4/] | 35, 36 | 9 | — | G | — | CLK | $AD_0$-$AD_{15}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL8}$ $t_{PLH8}$ | 116 [4/] | 43, 44 | 7,9 | — | | — | CLK | $A16/S_3$- $\overline{MEX}/S_7$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL9}$ $t_{PLH9}$ | 117 | | 7,9 | — | | — | CLK | $A16/S_3$- $\overline{MEX}/S_7$ | 10 | — | 10 | — | 10 | — | ns |
| $t_{PHL22}$ | 118 | | 8 | — | | — | $AD_0$-$AD_{15}$ $A16/S_3$-$A19/S_6$ | $\overline{RD}$ | 0 | — | 0 | — | 0 | — | ns |
| $t_{PHL10}$ | 119 | 21 | 8 | — | G | — | CLK | $\overline{RD}$ | 10 | 165 | 10 | 165 | 10 | 165 | ns |
| $t_{PLH10}$ | 120 | 22 | 8 | — | G | — | CLK | $\overline{RD}$ | 10 | 150 | 10 | 150 | 10 | 150 | ns |
| $t_{PHL11}$ | 121 [4/] | 31, 32 | 12 | — | G | — | CLK | $\overline{WR}/\overline{CS0}$, $\overline{WR}/\overline{CS1}$ | 0 | 85 | 0 | 85 | 0 | 85 | ns |
| $t_{PLH11}$ | 122 [4/] | 33 | 12 | — | G | — | CLK | $\overline{WR}/\overline{CS0}$, $\overline{WR}/\overline{CS1}$ | 0 | 85 | 0 | 85 | 0 | 85 | ns |
| $t_{PLH12}$ | 123 | 16 | 7,8,9 | — | G | — | CLK | ALE | 0 | 80 | 0 | 80 | 0 | 80 | ns |
| $t_{PHL12}$ | 124 | 9 | 7,8,9 | — | F | — | CLK | ALE | — | 85 | — | 85 | — | 85 | ns |
| $t_{PHL13}$ $t_{PLH13}$ | 125 | | 7,9 | — | | — | ALE | $\overline{MEX}/S_7$, $A15/S_3$-$A19/S_6$ | $t_{PHH1}$-10 | — | $t_{PHH1}$-10 | — | $t_{PHH1}$-20 | — | ns |

See footnotes at end of table.



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TABLE III C. Group A inspection - Continued.

| Symbol | Test number | Pattern number | Waveform figure | Parameter verified | Input cond. from table III A | Output cond. from table III B | Measurement terminal: Input under test | Measurement terminal: Output under test | Test limits 1/: Subgroup 9 $T_A$ = 25°C Min | Subgroup 9 $T_A$ = 25°C Max | Subgroup 10 $T_C$ = 125°C Min | Subgroup 10 $T_C$ = 125°C Max | Subgroup 11 $T_C$ = -55°C Min | Subgroup 11 $T_C$ = -55°C Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $t_{PLH23}$ | 126 | 23,24,25 | 8,9,10 | | G | --- | CLK | $\overline{WTA}$, $\overline{DEN}$, $\overline{OE}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PLH24}$ | 127 | 13 | 9 | | F | --- | CLK | $\overline{DEN}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL14}$ | 128 | | 7,8,9 | --- | | --- | $AD_0$-$AD_{15}$, $A19/B_6$ - $A16/B_3$ | ALE | $t_{PWL1}$-60 | --- | $t_{PWL1}$-60 | --- | $t_{PWL1}$-60 | --- | ns |
| $t_{PHL15}$ | 129 | | 9 | --- | | --- | $\overline{BHE}/B_7$ | ALE | $t_{PWL1}$-60 | --- | $t_{PWL1}$-60 | --- | $t_{PWL1}$-60 | --- | ns |
| $t_{PHL16}$ | 130 | 17,19 | 9,10 | --- | G | --- | CLK | $\overline{DT}$, $\overline{DEN}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL17}$ $t_{PLH17}$ | 131 | 8 | 7,8,10 | --- | F | --- | CLK | $M/\overline{IO}$ $DT/\overline{R}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHL18}$ $t_{PLH18}$ | 132 4/ | 19,20 | 13 | --- | G | --- | CLK | RLDA | 10 | 160 | 10 | 160 | 10 | 160 | ns |
| $t_{PHL19}$ $t_{PLH19}$ | 133 | | 9 | --- | | --- | ALE | $AD_0$-$AD_{15}$ | $t_{PWH1}$-10 | --- | $t_{PWH1}$-10 | --- | $t_{PWH1}$-10 | --- | ns |
| $t_{PHL21}$ $t_{PLH21}$ | 134 4/ | 14 | 12,13 | --- | G | --- | CLK | 2/ | $t_{PHL3}$ | 80 | $t_{PHL3}$ | 80 | $t_{PHL3}$ | 80 | ns |
| $t_{PHL20}$ | 135 | 10,11,12 | 8,9,10 | | F | --- | CLK | $\overline{DEN}$ | 10 | 110 | 10 | 110 | 10 | 110 | ns |
| $t_{PHZ1}$ $t_{PLZ1}$ | 136 4/ | 39 | 8,10 | --- | G | --- | CLK | $AD_0$-$AD_{15}$ | $t_{PHL3}$ | 80 | $t_{PHL3}$ | 80 | $t_{PHL3}$ | 80 | ns |
| $t_{PHZ2}$ $t_{PLZ2}$ | 137 | | 9 | --- | | --- | CLK | $AD_0$-$AD_{15}$ | 10 | --- | 10 | --- | 10 | --- | ns |
| $t_{PHZ3}$ $t_{PLZ3}$ | 138 | | 8 | --- | | --- | $AD_0$-$AD_{15}$ | ALE | $t_{PWH1}$-30 | --- | $t_{PWH1}$-30 | --- | $t_{PWH1}$-30 | --- | ns |
| $t_{PHZA}$ $t_{PLZA}$ | 139 | | 9 | --- | | --- | $AD_0$-$AD_{15}$ | $\overline{WR}$ | $t_{PWL1}$-10 | --- | $t_{PWL1}$-10 | --- | $t_{PWL1}$-10 | --- | ns |
| $t_{PZH}$ $t_{PZL}$ | 140 | | 8 | --- | | --- | $\overline{RD}$ | $AD_0$-$AD_{11}$ $AD_{16}/S_3$- $AD_{19}/S_6$ | $t_{CYC}$-45 | --- | $t_{CYC}$-45 | --- | $t_{CYC}$-45 | --- | ns |
| $V_{OL}$ $V_{OH}$ | 141 | 6 | 7,8,9,10, 11 | $t_{PWL1}$ | | --- | CLK | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V |

See footnotes at end of table.





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TABLE IIIC. Group A

| Symbol | Test number | Pattern number | Waveform figure | Parameter verified | Input cond. from table IIIA | Output cond. from table IIIB | Measurement terminal: Input under test | Measurement terminal: Output under test | Test limits 1/2/: Subgroup 9 $T_A$ = 25°C Min | Subgroup 9 $T_A$ = 25°C Max | Subgroup 10 $T_C$ = 125°C Min | Subgroup 10 $T_C$ = 125°C Max | Subgroup 11 $T_C$ = -55°C Min | Subgroup 11 $T_C$ = -55°C Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $t_{PHHA}$ | 145 | | 7,8,9 | --- | | | | ALE | $t_{PWL1}$-20 | --- | $t_{PWL1}$-20 | --- | $t_{PWL1}$-20 | --- | ns |
| $V_{OL}$ / $V_{OH}$ | 146 | 6 | 7,8,9,10, 11 | $t_{CTC}$ | | | CLK | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 147 | 6 | 7 | $t_{SRL1}$ | H | C,E 6/ | READY | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 148 | | 12 | $t_{SRL2}$ | I | D,F 6/ | $\overline{RQ}/GT0$ / $\overline{RQ}/GT1$ | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 149 | | 13 | $t_{SRL3}$ / $t_{SLH3}$ | H | C,E 6/ | HOLD | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 150 | | 14 | $t_{SRLA}$ / $t_{SLH4}$ | H | C,E 6/ | BH1 / INTR / $\overline{TEST}$ | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 151 | | 7,11 | $t_{SLH1}$ | | | READY | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 152 | | 8,10 | $t_{SEH}$ / $t_{SEL}$ | H | C,E 6/ | $AD_0$-$AD_{15}$ | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 153 | | 7,11 | $t_{SHL1}$ / $t_{SLH1}$ | | | READY | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 154 | | 12 | $t_{SLH2}$ | | | $\overline{RQ}/GT0$ / $\overline{RQ}/GT1$ | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |
| $V_{OL}$ / $V_{OH}$ | 155 | | 8,10 | $t_{SRH}$ / $t_{SLH}$ | H | C,E 6/ | $AD_0$-$AD_{15}$ | All outputs | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | V / V |

1/ Subgroups 9, 10, and 11 are performed with $V_{CC}$ = 4.5 V and 5.5 V.

2/ Subgroup 9, patterns 7-15 are applied with input condition F from table IIIA. Patterns 16-64 are applied with input condition G from table IIIA.

3/ The capacitance measurements shall be made between the indicated terminal and ground at a frequency of 1 MHz. The d.c. bias of the measuring instrument shall be less than ±0.1 V. The a.c. signal amplitude shall be less than 50 mVrms.

4/ Minimum delay times tested during initial qualification and after process or design changes.

5/ Where propagation delay tests for subgroup 9 indicate a group of pins and where two or more propagation delays are specified for a test, only the reading for the pin with the worst case delay requires recording. In some cases this will be a minimum reading while for others it will be a maximum reading as indicated in the table.

6/ Clock is pulsed continuously between 0.0 and 4.0 V.

7/ This delay is for outputs $AD_0$-$AD_{15}$, $A19/S_6$-$A16/S_3$, $\overline{S0}$-$\overline{S2}$, $\overline{BHE}$, $\overline{LOCK}$, and $\overline{SSR}/S_7$ during request/grant and HOLD/HOLDA sequences only.

8/ Output conditions C and D are used for subgroup 9, E and F are used for subgroups 10 and 11.





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TABLE IV. Vectors.

Table IV, not printed herein, is a computer listing for use with table IIIC. Each vector identifies activity states by pin number. See 6.2 for information on obtaining the complete table in compatible output format.

NOTES:

The ac, dc and functional test vectors are divided into 44 patterns which are defined as:

| Subgroup | Pattern | Type | Vector number |
|---|---|---|---|
| 1,2,3 | 1 | D.C. | 0001-1497 |
| ↓ | 2 | ↓ | 1498-2994 |
| | 3 | | 2995-4492 |
| | 4 | | 4493-4506 |
| ↓ | 5 | ↓ | 4507-4524 1/ |
| 7,8,9 | 6 | Functional A.C. | 4582-22467 |
| 9,10,11 | 7 | A.C. | 22468-23503 |
| ↓ | 8 | ↓ | 23504-25694 |
| | 9 | | 25695-26731 |
| | 10 | | 26732-27784 |
| | 11 | | 27785-28925 |
| | 12 | | 28926-29963 |
| | 13 | | 29964-31019 |
| | 14 | | 31020-31932 |
| | 15 | | 31933-33417 |
| | 16 | | 33418-34448 |
| | 17 | | 34449-35501 |
| | 18 | | 35502-36641 |
| | 19 | | 36642-37665 |
| | 20 | | 37666-38692 |
| | 21 | | 38693-39729 |
| | 22 | | 39730-40768 |
| | 23 | | 40769-41807 |
| | 24 | | 41808-42862 |
| | 25 | | 42863-44004 |
| | 26 | | 44005-44919 |
| | 27 | | 44920-44961 |
| | 28 | | 44962-45005 |
| | 29 | | 45006-45919 |
| | 30 | | 45920-46840 |
| | 31 | | 46841-46868 |
| | 32 | | 46869-46899 |
| | 33 | | 46900-46931 |
| | 34 | | 46932-46999 |
| | 35 | | 47000-48247 |
| | 36 | | 48248-49513 |
| | 37 | | 49514-50764 |
| | 38 | | 50765-52033 |
| | 39 | | 52034-53179 |
| | 40 | | 53180-53216 |
| | 41 | | 53217-54471 |
| | 42 | | 54472-55075 |
| | 43 | | 55076-55253 |
| ↓ | 44 | ↓ | 55254-56509 |

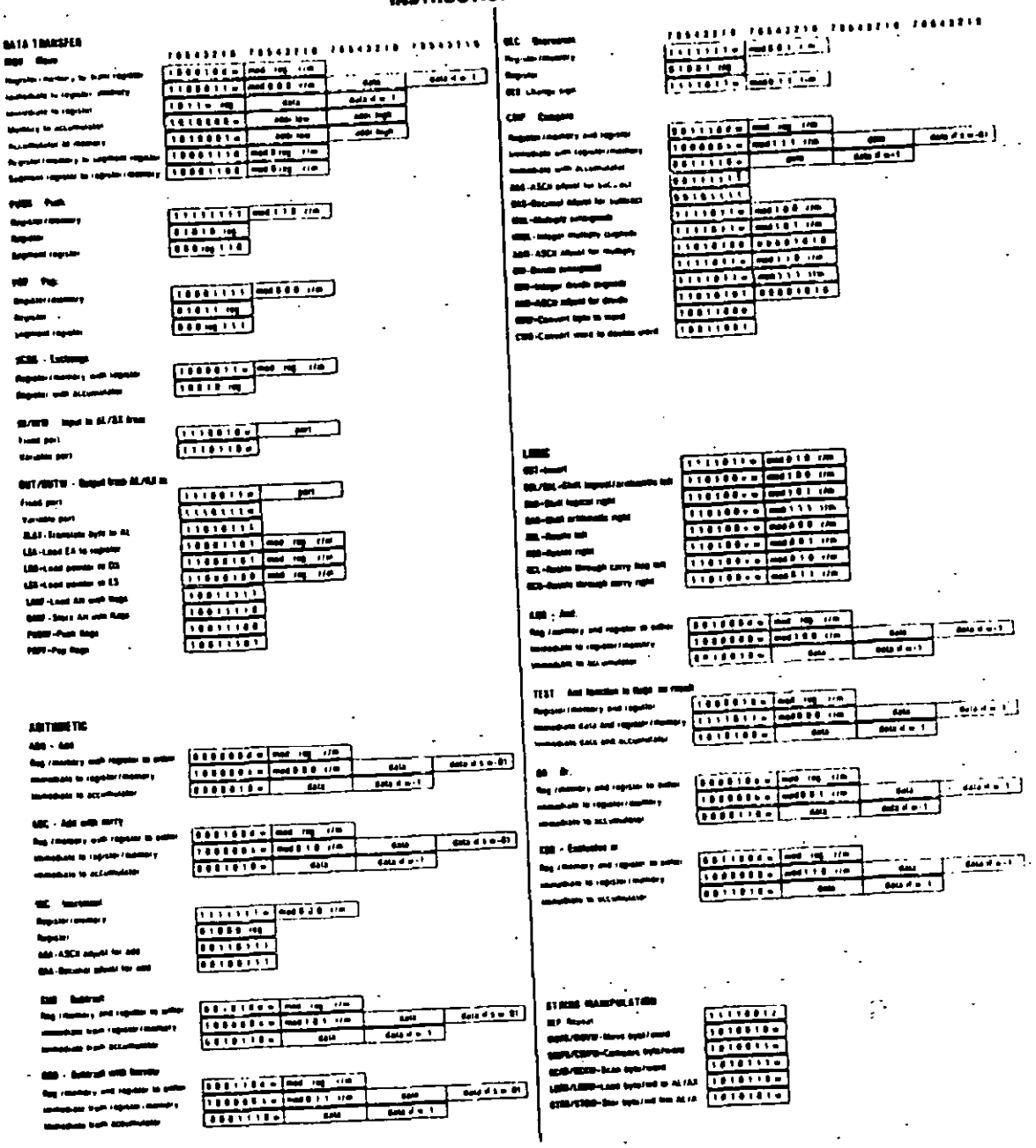
1/ Vectors 4525-4581 are omitted here but do appear in table IIIC as measurement tests for $C_I$, $C_O$, $I_{ZL}$, and $I_{ZH}$.

2/ Z = High impedance = 1.5 V.
X = Don't care

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TABLE V.

INSTRUCTION SET SUMMARY

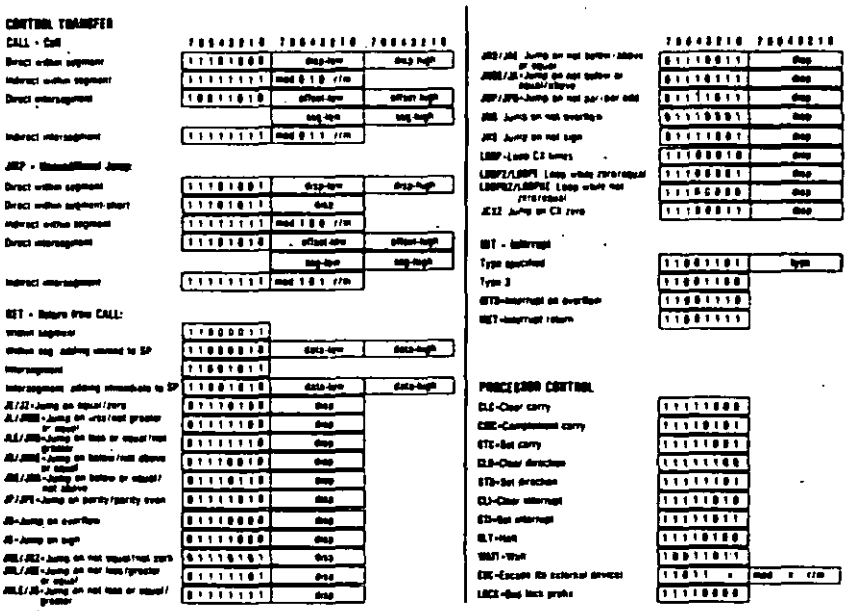




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TABLE V.

INSTRUCTION SET SUMMARY , Continued

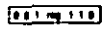


Footnotes:

AL = 8-bit accumulator
AX = 16-bit accumulator
CX = Count register
DS = Data segment
ES = Extra segment
Above/below refers to unsigned value
Greater = more positive.
Less = less positive (more negative) signed values
if d = 1 then "to"; if d = 0 then "from"
if w = 1 then word instruction; if w = 0 then byte instruction

if s w = 01 then 16 bits of immediate data form the operand
if s w = 11 then an immediate data byte is sign extended to form the 16-bit operand
if v = 0 then "count" = 1; if v = 1 then "count" in (CL)
x = don't care
z is used for string primitives for comparison with ZF FLAG

SEGMENT OVERRIDE PREFIX



if mod = 11 then r/m is treated as a REG field
if mod = 00 then DISP = 0*, disp-low and disp-high are absent
if mod = 01 then DISP = disp-low sign-extended to 16-bits, disp-high is absent
if mod = 10 then DISP = disp-high, disp-low

if r/m = 000 then EA = (BX) + (SI) + DISP
if r/m = 001 then EA = (BX) + (DI) + DISP
if r/m = 010 then EA = (BP) + (SI) + DISP
if r/m = 011 then EA = (BP) + (DI) + DISP
if r/m = 100 then EA = (SI) + DISP
if r/m = 101 then EA = (DI) + DISP
if r/m = 110 then EA = (BP) + DISP*
if r/m = 111 then EA = (BX) + DISP

DISP follows 2nd byte of instruction (before data if required)

*except if mod = 00 and r/m = 110 then EA = disp-high, disp-low

REG is assigned according to the following table

| 16-Bit (w = 1) | 8-Bit (w = 0) | Segment |
|---|---|---|
| 000 AX | 000 AL | 00 ES |
| 001 CX | 001 CL | 01 CS |
| 010 DX | 010 DL | 10 SS |
| 011 BX | 011 BL | 11 DS |
| 100 SP | 100 AH | |
| 101 BP | 101 CH | |
| 110 SI | 110 DH | |
| 111 DI | 111 BH | |

Instructions which reference the flag register file as a 16-bit object use the symbol FLAGS to represent the file

FLAGS = X X X X (OF) (DF) (IF) (TF) (SF) (ZF) X (AF) X (PF) X (CF)



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g. Requirement for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct shipment to the Government.

h. Requirement for "JAN" marking.

6.5 Terms and definitions. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

2-16, 39 — Address data bus: These lines constitute the time multiplexed memory/$I_O$ address ($T_1$) and data ($T_2$, $T_3$, $T_W$, $T_4$) bus. $A_0$ is analogous to $\overline{BHE}$ for the lower byte of the data bus, pins $D_7$ - $D_0$. It is LOW during $T_1$ when a byte is to be transferred on the lower portion of the bus in memory or I/O operations. Eight-bit oriented devices tied to the lower half would normally use $A_0$ to condition chip select functions. (See $\overline{BHE}$). Those lines are active HIGH and float to 3-state OFF during interrupt acknowledge and local bus "hold acknowledge".

35-38 — Address/status: During $T_1$ these are the four most significant address lines for memory operations. During I/O operations these lines are LOW. During memory and I/O operations, status information is available on these lines during $T_2$, $T_3$, $T_W$, and $T_4$. The status of the interrupt enable FLAG bit ($S_5$) is updated at the beginning of each CLK cycle. $A_{17}/S_4$ and $A_{16}/S_3$ are encoded as shown. This information indicates which relocation register is presently being used for data accessing.

These lines float to 3-state OFF during local bus "hold acknowledge."

| $A_{17}/S_4$ | $A_{16}/S_3$ | Characteristics |
|---|---|---|
| 0(LOW) | 0 | Alternate Data |
| 0 | 1 | Stack |
| 1(HIGH) | 0 | Code or None |
| 1 | 1 | Data |
| $S_6$ is 0 (LOW) | | |

34 — Bus high enable/status: During $T_1$ the bus high enable signal ($\overline{BHE}$) should be used to enable data onto the most significant half of the data bus, pins $D_{15}$-$D_8$.

$\overline{BHE}$ is LOW during $T_1$ for read, write, and interrupt acknowledge cycles when a byte is to be transfered on the high portion of the bus. The $S_7$ status information is available during $T_2$, $T_3$, and $T_4$. The signal is active LOW, and floats to 3-state OFF in "hold". It is LOW during $T_1$ for the first interrupt acknowledge cycle.

| $\overline{BHE}$ | $A_0$ | Characteristics |
|---|---|---|
| 0 | 0 | Whole word |
| 0 | 1 | Upper byte from/ to odd address |
| 1 | 0 | Lower byte from/ to even address |
| 1 | 1 | None |



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| | |
|---|---|
| 32 | Read: Read strobe indicates that the processor is performing a memory or I/O read cycle, depending on the state of the $S_2$ pin. This signal is used to read devices which reside on the local bus. $\overline{RD}$ is active LOW during $T_2$, $T_3$, and $T_W$ of any read cycle, and is guaranteed to remain HIGH in $T_2$ until the 8086 local bus has floated.<br><br>This signal floats to 3-state OFF in "hold acknowledge". |
| 22 | Ready: Ready is the acknowledgement from the addressed memory or I/O device that it will complete the data transfer. This signal is active HIGH. The Ready input is not synchronized. |
| 18 | Interrupt request: Interrupt request is a level triggered input which is sampled during the last clock cycle of each instruction to determine if the processor should enter into an interrupt acknowledge operation. It can be internally masked by software resetting the interrupt enable bit. INTR is internally sychronized. This signal is active HIGH. |
| 23 | Test: Test input is examined by the "Wait" instruction. If the $\overline{TEST}$ input is LOW execution continues, otherwise the processor waits in an "Idle" state. This input is synchronized internally during each clock cycle on the leading edge of CLK. |
| 17 | Non-maskable interrupt: An edge triggered input which causes a type 2 interrupt. A subroutine is vectored to via an interrupt vector lookup table located in system memory. NMI is not maskable internally by software. A transition from a LOW to HIGH initiates the interrupt at the end of the current instruction. This input is internally synchronized. |
| 21 | Reset: Reset causes the processor to immediately terminate its present activity. The signal must be active HIGH for a least four clock cycles. It restarts execution, as described in the Instruction Set description, when RESET returns LOW. Reset is internally synchronized. |
| 19 | Clock: Clock provides the basic timing for the processor and bus controller. It is asymmetric with a 33 percent duty cycle to provide optimized internal timing. |
| 40 | $V_{CC}$: + 5 V power supply pin. |
| 1,20 | Ground |
| 33 | Minimum/maximum: Indicates what mode the processor is to operate in. The two modes are discussed in the following sections:<br><br>Pins 24-31, when operating in maximum mode, perform the following function: |
| 26-28 | Status: Active during $T_4$, $T_1$, and $T_2$ and is returned to the passive state (1,1,1) during $T_3$ or during $T_W$ when READY is HIGH. This status is used by the Bus Controller to generate all memory and I/O access control signals. Any change by $\overline{S_2}$, $\overline{S_1}$, or $\overline{S_0}$ during $T_4$ is used to indicate the beginning of a bus cycle, and the return to the passive state in $T_3$ or $T_W$ is used to indicate the end of a bus cycle.<br><br>These signals float to 3-state OFF in "hold acknowledge". These status lines are encoded as shown. |

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| $\overline{S_2}$ | $\overline{S_1}$ | $\overline{S_0}$ | Characteristics |
|---|---|---|---|
| 0(LOW) | 0 | 0 | Interrupt Acknowledge |
| 0 | 0 | 1 | Read I/O Port |
| 0 | 1 | 0 | Write I/O Port |
| 0 | 1 | 1 | Halt |
| 1(HIGH) | 0 | 0 | Code Access |
| 1 | 0 | 1 | Read Memory |
| 1 | 1 | 0 | Write Memory |
| 1 | 1 | 1 | Passive |

30,31 Request/grant: Pins are used by other local bus masters to force the processor to release the local bus at the end of the processor's curent bus cycle. Each pin is bidirectional with $RQ/GT_0$ having higher priority than $RQ/GT_1$. RQ/GT has an internal pull-up resistor so may be left unconnected. The request/grant sequence is as follows (see figure 9):

1. A pulse of 1 CLK wide from another local bus master indicates a local bus request ("hold") to the (pulse 1).

2. During the CPU's next $T_4$ or $T_1$ a pulse 1 CLK wide from the processor to the requesting master (pulse 2), indicates that the processor has allowed the local bus to float and that it will enter the "hold acknowledge" state at the next CLK. The CPU's bus interface unit is disconnected logically from the local bus during "hold acknowledge".

3. A pulse 1 CLK wide from the requesting master indicates to the processor (pulse 3) that the "hold" request is about to end and that the processor can reclaim the local bus at the next CLK.

Each master-master exchange of the local bus is a sequence of 3 pulses. There must be one dead CLK cycle after each bus exchange. Pulses are active LOW.

29 LOCK: Output indicates that other system bus masters are not to gain control of the system bus while LOCK is active LOW. The LOCK signal is activated by the "LOCK" prefix instruction and remains active until the completion of the next instruction. This signal is active LOW, and floats to 3-state OFF in "hold acknowledge".

24-25 Queue status: The queue status is valid during the CLK cycle after which the queue operation is performed.

$QS_1$ and $QS_0$ provide status to allow external tracking of the internal instruction queue.

Pins 24-31, when operating in minimum mode, perform the following function:

28 Status line: Logically equivalent to $S_2$ in the maximum mode. It is used to distinguish a memory access from an I/O access. M/IO becomes valid in the $T_4$ preceding a bus cycle and remains valid until the final $T_4$ of the cycle (M = HIGH, IO = LOW). M/IO floats to 3-state OFF in the local bus "hold acknowledge".

29 Write: Indicates that the processor is performing a write memory or write I/O cycle, depending on the state of the M/IO signal. WR is active for $T_2$, $T_3$, and $T_W$ of any write cycle. It is active LOW, and floats to 3-state OFF in local bus "hold acknowledge."



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24 INTA: Is used as a read strobe for interrupt acknowledge cycles. It is active LOW during $T_2$, $T_3$, and $T_W$ of each interrupt acknowledge cycle.

25 Address latch enable: Provided by the processor to latch the address into the address latch. It is a HIGH pulse active during $T_1$ of any bus cycle. Note that ALE is never floated.

27 Data transmit/receive: Needed in minimum system that desires to use a data bus transceiver. It is used to control the direction of data flow through the transceiver. It is used to control the direction of data flow through the transceiver. Logically DT/R is equivalent to $S_1$ in the maximum mode, and its timing is the same as for M/IO. (T = HIGH, R = LOW). this signal floats to 3-state OFF in local bus "hold acknowledge".

26 Data enable: Provided as an output enable for a minimum system which uses the transceiver. DEN is active LOW during each memory and I/O access and for INTA cycles. For a read or INTA cycle it is active from the middle of $T_2$ until the middle of $T_4$, while for a write cycle it is active from the beginning of $T_2$ until the middle of $T_4$. DEN floats to 3-state OFF in local bus "hold acknowledge".

31 HOLD: Indicates that another master is requesting a local bus "hold". To be acknowledged. HOLD must be active HIGH. The processor receiving the "hold" request will issue HLDA (HIGH) as an acknowledgement in the middle of $T_4$ or $T_1$. Simultaneous with the issuance of HLDA the processor will float the local bus and control lines. After HOLD is detected as being LOW, the processor will LOWer HLDA, and when the processor needs to run another cycle, it will again drive the local bus and control lines.

HOLD is not an asynchronous input. External synchronization should be provided if the system cannot otherwise guarantee the setup time.

6.6 Logistic support. Lead material and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), and lead material and finish "C" (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.7 Handling. MOS microprocessors should be handled with certain precautions to avoid damage due to the discharge of accumulated static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage, however, the following handling practices are recommended.

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

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6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

| Military-device type | Generic-industry type |
|---|---|
| 01 | 8086 |

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-F513)



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